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Order Flow, Liquidity, and Security Price Dynamics

Learning Objectives:

- Why and how orders move prices
- Why there is a bid-ask spread
- How prices are formed when orders convey information
- The determinants of market illiquidity
- What inventory risk is and how it affects prices

3.1 Introduction

In principle, asset prices change in response to news about fundamentals, namely, future cash flows and discount factors. But security prices fluctuate continuously, even within extremely short time spans and in the absence of price-relevant news. These movements are responses to incoming orders to buy or sell stocks. Market microstructure theory explains how these intraday price movements are related to the order flow. It identifies illiquidity as one cause of these short-run fluctuations, which can be either transient or longer lasting depending on the motives underlying orders and the way market participants interpret them. If orders are seen as reflecting news about fundamentals, the price movements tend to be long lasting; otherwise, they tend to be reversed quickly.

The prevalence of very short-term price movements and their responsiveness to orders is illustrated in Table 3.1, which reports trade and quote data for the first transactions of a typical trading day (March 26, 2001) for the stock of a large French insurance company, AGF.¹ This stock was traded on Euronext's electronic limit order market until it was acquired by Allianz in 2007. The type of data

¹ The companion website for this book reports the same data for the entire day (519 trades).

Table 3.1 Sample Trading Session for AGF on Euronext, March 26, 2001

Time	Trade size	Price	Direction	Bid	Ask
t	(q_t)	(p_t)	(d_t)	(b_t)	(a_t)
9:06:04	20	66.70	-1	66.90	67.00
9:06:11	25	66.64	-1	66.65	66.70
9:06:26	18	66.60	-1	66.60	66.65
9:07:18	273	66.42	-1	66.50	66.55
9:07:36	27	66.55	+1	66.15	66.55
9:18:03	100	66.25	-1	66.25	66.35
9:19:37	267	66.20	-1	66.20	66.30
9:23:08	12	66.15	-1	66.15	66.25
9:23:31	157	66.15	-1	66.15	66.20
9:23:38	30	66.10	-1	66.10	66.15
9:26:26	1,000	66.20	+1	66.00	66.20
9:30:10	1	66.20	+1	66.05	66.20
9:30:54	24	66.16	+1	66.05	66.20
9:34:40	90	66.05	-1	66.05	66.20
9:35:39	6	66.05	-1	66.05	66.20
9:36:10	1,000	66.00	-1	66.00	66.20
9:36:14	15	66.20	+1	66.00	66.20
9:39:56	75	66.05	-1	66.05	66.10

given in Table 3.1 illustrates well the type of high-frequency data available on transactions in many markets (see exercise 11 for instance).

For each transaction between 9:05 a.m. and 9:40 a.m., the table displays: (i) the time of the trade, t ; (ii) its absolute size, $|q_t|$; (iii) the price, p_t , at which it is executed; (iv) the highest bid quote, b_t , and the lowest ask quote a_t on the limit order book immediately before the trade; and (v) the direction, d_t , of the trade, defined as +1 or -1 depending on whether the party initiating the trade (i.e., demanding liquidity) is buyer or seller. The (signed) *order flow* is thus the absolute trade size multiplied by its direction: $q_t = |q_t| \cdot d_t$.

Clearly, the price varies from one transaction to the next.² There is considerable movement even though the time interval is just thirty-four minutes, and this pattern is in no way specific to this particular stock or period. This chapter serves precisely to provide a framework for understanding such intraday stock

² The transaction price is not necessarily equal to the bid or the ask price just before the transaction. In a limit order market, the depth at the best quotes may be too little to fully execute a marketable order. In this case, the order walks up or down the book (depending on its direction). So the average transaction price can be higher than the current ask or lower than the current bid price. The price can also be strictly inside the quotes if it is a matched trade pre-negotiated outside the main market and entered into the system by a special procedure. For a description of the trading operations in a limit order market, see Chapter 1; for a formal analysis of this trading mechanism, see Chapter 6.

price variations. This framework also helps to explain the factors that determine the other variables appearing in Table 3.1, in particular the order flow and the bid-ask spread. It will also clarify how price volatility, spreads, and order flow are interrelated. Lastly, it will lay the ground work for a more detailed, subsequent analysis of market design issues.

We start with the Efficient Market Hypothesis (EMH), a pillar of modern finance theory, according to which security price changes are induced by the arrival of new information and should follow a random walk. In Section 3.2 we briefly review the EMH to observe that, while useful as a starting point, it does not exactly address the way information gets embodied in asset prices; in particular, it does not assign any explicit role to the variables reported in Table 3.1, such as the bid and ask quotes and the order flow. This is problematic, as price changes do appear to relate to these variables. Specifically, when markets are not perfectly liquid (that is, when the spread is not zero), buy market orders push prices upward, and sell market orders push them downward. The larger the order, the sharper is the price movement, and price volatility is related to the order flow.

These empirical patterns can be reconciled with the EMH by recognizing that the order flow itself provides new information. A series of buy orders is a signal that informed traders may be buying because the stock is undervalued, just as a series of sell orders suggests it may be overvalued. Accordingly, liquidity suppliers should revise their expectations concerning the stock's value based on the observed order flow, and price changes should be partly determined by the order flow.

Section 3.3 sets out a trading model that formalizes this intuition. The model yields several insights. For instance, it explains the existence of the bid-ask spread and relates price changes to the order flow. In this model, the bid-ask spread is a compensation required by liquidity suppliers to offset their potential losses on trades with better informed investors. Thus, the asymmetry of information between the suppliers and the demanders of liquidity determines the market's degree of liquidity.

But in reality, liquidity is also determined by other factors. In particular, liquidity suppliers need to cover the cost of executing orders and maintaining a presence in the market (so-called order-processing costs). Moreover, imperfect competition among them generates rents, which take the form of bid-ask spreads greater than order-processing costs. In Section 3.4, we enrich the model to include both order-processing costs and the rents due to imperfect competition among liquidity suppliers.

Another reason for the existence of a bid-ask spread is the fact that dealers are risk averse. As prices fluctuate during the trading day, liquidity suppliers are exposed to variations in the value of their positions ("inventory risk"),

because these positions cannot necessarily be unwound immediately. As a dealer accumulates a long position in the stock, he is exposed to the risk of a price drop; conversely, a short position entails the danger of a rise in the price. If the dealer is risk averse, his quotes should depend directly on his exposure to inventory risk. Section 3.5 analyzes the impact of inventory risk and dealers' risk aversion on bid and ask quotes.

Overall, the models presented in this chapter describe three kinds of cost for liquidity suppliers: (i) the cost of trading with better informed investors (adverse selection costs), (ii) the real cost of processing orders (order-processing costs), and (iii) the cost of holding risky assets (inventory holding costs). Since liquidity suppliers need a larger bid-ask spread to compensate for these costs, each cost category will contribute to market illiquidity. In addition, oligopolistic liquidity suppliers may charge a markup that also fosters illiquidity. Understanding the relative importance of these determinants of liquidity is important for policy-making and market design. For instance, if order-processing costs predominate, then a change in the trading system might be called for. But if adverse selection costs are the main factor, improvements in disclosure and greater parity between traders should be considered.

Thus, it is important to distinguish between the various costs borne by liquidity suppliers. The models presented in this chapter suggest a way to assess their relative importance, because they show that different costs carry different implications for short-term price movements after the execution of a market order. Even though the immediate impact of a market order is the same in all cases (a buy pushes the price up; a sell, down), the subsequent price dynamics differ with the source of market illiquidity, as explained here (summarized in Section 3.6). Specifically, if the bid-ask spread is compensation for adverse selection costs, then the price impact of a market order is permanent, as it leads liquidity suppliers to revise their estimate of the security's value. But if the spread is compensation for order-processing costs or inventory risk, then the price impact of a market order is transient and should dissipate over time; that is, these kinds of cost induce reversals in returns. Moreover, the speed of the reversal differs depending on the relative importance of inventory holding vis-à-vis order-processing costs.

Casual observation of price movements during the day suggests that the immediate price impact of market orders actually has both a permanent and a transient component: in the real world, all kinds of costs matter. Chapter 5 describes various empirical techniques to exploit the implications of the models presented here for assessing the relative importance of adverse-selection costs, order-processing costs, and inventory holding costs.

The analysis in much of this chapter is based on the simplifying assumption that all orders are for a fixed number of shares, which we normalize to 1 so

that $q_t = d_t \in \{-1, +1\}$.³ Trades occur when investors (“liquidity demanders”) submit orders to buy or sell; the trades are registered in the sequence in which they occur, just as in Table 3.1. Thus, p_t is the price at which the t^{th} trade takes place, and d_t indicates whether the investor initiating the t^{th} trade is a buyer or a seller. Liquidity is supplied by a pool of traders, whom we call “dealers” or “market makers,” but who could also be limit order traders: in this framework, the precise nature of those providing liquidity is irrelevant. Dealers post their bid and ask quotes, at which they are willing to execute sell or buy orders. We denote by b_t^i the bid price posted by dealer i at time t (i.e., what dealer i bids for the stock). Similarly, we denote by a_t^i the ask (or offer) price posted by dealer i at time t (i.e., the price that he asks for the security or at which he offers it for sale). We denote the best ask price by a_t and the best bid price by b_t :

$$a_t = \min_i \{a_t^i\} \quad \text{and} \quad b_t = \max_i \{b_t^i\}.$$

Investors trade with the dealer that posts the best price: they buy at the lowest ask price a_t and sell at the highest bid price b_t . So the transaction price at time t is either a_t or b_t , depending on the direction of the order that triggers it.

3.2 Price Dynamics and the Efficient Market Hypothesis

Some football clubs are listed on stock exchanges. Not surprisingly, their stock price is very sensitive to their on-field performance: a good showing brings more fans and television coverage, swelling expected revenues. So the outcome of matches affects the club’s stock market value. Most matches take place on weekends. On Monday, the stocks of winners rise typically and those of losers fall (Palomino et al. 2009). This illustrates a general principle: new information is a source of price changes. For any security, new information leads investors to revise their estimates of future cash flows. The price adjusts to reflect investors’ updated valuation.

The speed of this adjustment is an important issue. For instance, imagine that Ajax Amsterdam meets Juventus in the UEFA Champions League, and Ajax wins in an upset. Assume the closing price for Ajax on the last trading day before the match was €7. In view of Ajax’s victory, investors revise their estimate of its stock value (i.e., the discounted value of its expected future dividends) to €7.20. How does its stock price evolve on the trading day after the match? One scenario

³ In Section 3.5, and more extensively in Chapter 4, we will allow investors to place orders of varying size.

could be that it will gradually approach its new fundamental level of €7.20. For instance, a first trade takes place at $p_1 = €7.10$, a second at $p_2 = €7.15$, and so on.

The EMH, however, holds that the adjustment should be instantaneous.⁴ According to the EMH, at any point in time, trades are made at a price that is equal to the best possible estimate of the value of the asset, incorporating all available information. We shall refer to this as the “fundamental value” v of the security, which can be thought of as the value once trading is over. Formally, the EMH states that:

$$p_t = \mu_t \equiv E(v|\Omega_t), \quad (3.1)$$

where μ_t is the market makers’ estimate of the security’s value v as of time t , and Ω_t denotes the information available to them at that time. The notation $E(v|\Omega_t)$ indicates the expected value of v conditional on information Ω_t . The absence of a discount factor is justified by the shortness of the interval (intraday).

The conditional expectation μ_t can change only if new information reaches the market. Let the random variable $\varepsilon_{t+1} = \mu_{t+1} - \mu_t$ represent the revision in investors’ value estimates induced by the news arriving between time t and $t + 1$, also known as the “innovation” in value. As the innovation at any time captures the effect of news, it cannot be forecast using past information: otherwise, news would not really be news! This means that $E(\varepsilon_t \varepsilon_s) = 0$, for $s \neq t$ (otherwise past innovations could be used to forecast future ones), and that $E[\varepsilon_{t+1} | \Omega_t] = 0$, which implies that the expectation of μ_{t+1} as of time t is simply μ_t :

$$E[\mu_{t+1} | \Omega_t] = \mu_t. \quad (3.2)$$

As $p_t = \mu_t$ at each date, it follows immediately that:

$$p_t = E(p_{t+1} | \Omega_t). \quad (3.3)$$

Equation (3.3) carries an important implication for the dynamics of stock prices: namely, that the best predictor of future prices, given currently available information, is the current price. In other words, under the EMH transaction prices follow a *martingale*: price changes over a given interval are serially uncorrelated. This can be seen by computing the change of p_t from equation (3.1):

⁴ Palomino et al. (2009) actually find some evidence that contradicts the EMH: the stock price does not seem to adjust immediately to bad news, or to incorporate all relevant public information.

$$\Delta p_{t+1} = p_{t+1} - p_t = \mu_{t+1} - \mu_t = \varepsilon_{t+1}, \quad (3.4)$$

so that $\text{cov}(\Delta p_t, \Delta p_s) = \text{cov}(\varepsilon_t, \varepsilon_s) = 0$, since $E(\varepsilon_t \varepsilon_s) = 0$, for $s \neq t$.

Under the EMH, what drives intraday price changes is new information, and assets' prices adjust immediately to the fundamental values given the available information. Hence, any change in price must be due to completely unanticipated information, so that changes cannot be predicted from past information, and particularly from previous price changes.

Let us return to the football club example. On the trading day after the match with Juventus, Ajax opens at €7.20. According to the EMH all subsequent trades should be at this price as long as no new information about Ajax arrives. But suppose that later Ajax announces that its goalkeeper was injured during the match and will be unable to play in future matches. If this worsens Ajax's prospects for the rest of the season, traders mark down their expectations regarding the value of Ajax ($\varepsilon_{t+1} < 0$), and the price immediately drops by $|\varepsilon_{t+1}|$.

The question, however, is how markets become informationally efficient. In other words, what is the *process* whereby equation (3.1) comes to hold? To answer, we need to be more specific about the details of the trading process and the nature of the information that market participants have. Suppose, for instance, that (i) dealers are risk neutral and competitive, (ii) investors do not have more information than dealers, and (iii) trading is cost free. Assumption (ii) implies that the order flow does not contain information; that is, that innovations ε_t and the direction-of-trade indicator, d_t , are independent. If so, dealers have no reason to update their value estimate when they receive buy or sell orders.

Under these circumstances, there is a zero bid-ask spread, and prices do not change in response to orders. To see this, consider how a dealer i sets his ask price at time t , a_t^i . If he sells the stock at price a_t^i , he obtains an expected profit equal to:

$$E[(a_t^i - v) | \Omega_t] = a_t^i - \mu_t.$$

Thus, the lowest price at which the dealer is willing to sell is his estimate of the stock's value, μ_t . Since this is the case for all market makers, competition drives their ask price to μ_t . A similar argument shows that the bid price is also μ_t . Thus, dealers offer to execute all orders at price μ_t , the bid-ask spread is nil, and any price movement is entirely attributable to the arrival of public information, not to the order flow.

This is a useful benchmark model, but it fails to capture some important aspects of the intraday trading process. First, empirical studies have shown that intraday price volatility is too great to be explained solely by news (French and Roll 1986 and Roll 1988). This suggests that the trading process itself is a source

of volatility. Second, the model fails to capture the simple fact that positive bid-ask spreads are the norm ($a_t - b_t > 0$). Lastly, in practice, intraday changes in prices are often negatively correlated. For instance, for the data reported in Table 3.1, the correlation between successive price changes is negative (-0.45), in line with the empirical evidence summarized by Stoll (2000).⁵

In the rest of this chapter, we show that these different features of the intraday trading process can be captured by relaxing assumptions (i), (ii), and (iii), which is to say, by introducing *frictions*. As we shall see, then, it is no longer the case that the transaction price p_t is equal to the dealers' estimate of the asset value, μ_t . What does determine the price changes between one transaction and the next, Δp_t , and their volatility, $\text{var}(\Delta p_t)$? What are the determinants of the bid and ask quotes, a_t and b_t ? What is the relationship between price changes and order flow, q_t ? Are changes in prices still serially uncorrelated? We now turn to these questions.

3.3 Price Dynamics with Informative Order Flow

We have seen that under the EMH the price of securities reflects investors' beliefs about these securities' fundamental values, which are continuously updated on the basis of new information, as captured by the term ε_t in equation (3.4). Where does this information come from? It could be generated by public announcements about company performance and macroeconomic news (say, interest rate changes). For instance, prices react to companies' announcements of dividends, insofar as the announcement is really a surprise. But not all information is broadcast simultaneously to all market participants: some investors may be privy to price-relevant information before other market participants, possibly thanks to tips from the company's management or because they have superior analytical skills.⁶

When these investors trade on their private knowledge, their orders convey information to the rest of the market, over and above what is already publicly available. In this environment, market participants will revise their estimate of securities' values in light of the order flow: unusual buying pressure will induce price increases; unusual selling pressure, price declines.

⁵ There is also considerable debate among scholars on whether the EMH is a good description of price formation at lower frequencies, for instance, monthly or even yearly: see the overview by Shleifer (2000).

⁶ Insiders (i.e., employees of a firm, board members, or people with close connections to them) are very well placed to obtain private information. Insider trading is strictly regulated (e.g., insiders cannot trade in the run-up to major company announcements) precisely in order to mitigate the risk of informed trading in the marketplace. Nevertheless, cases of illegal insider trading are frequent and constitute a good example of informed trading.

This feature can explain the existence of a bid-ask spread. To understand this point intuitively, consider the following argument, first presented by Jack Treynor in 1971 under his pseudonym “Bagehot.” In principle, dealers are not always as well informed as each and every customer. Traders with superior information will exploit any mispricing by dealers, buying when the ask price is lower than the fundamental value and selling when the bid price is higher. Dealers lose money when they trade with such investors. This is known as “adverse selection”: due to the informational asymmetry, market makers tend to attract customers who expect to make a profit at the dealer’s expense.⁷ To recoup their losses on informed orders, dealers must gain on their business with other traders. They achieve this by means of the bid-ask spread vis-à-vis all customers.

We develop these ideas in the context of a simplified version of the model proposed by Glosten and Milgrom (1985), a cornerstone of empirical and theoretical developments in this field.⁸ In this model the ask price exceeds the bid because the former is set in anticipation of receiving a buy order, the latter a sell order:

$$\begin{aligned}a_t &= E(v | \Omega_{t-1}, d_t = +1), \\b_t &= E(v | \Omega_{t-1}, d_t = -1),\end{aligned}\tag{3.5}$$

where Ω_{t-1} is the information known right up to the last trade d_t (recall that in most of this chapter $q_t = d_t$, since order sizes are normalized to 1). A positive bid-ask spread arises because of the informational content of the order flow d_t , which leads dealers to see orders as a source of information additional to that previously available, Ω_{t-1} .

3.3.1 The Glosten-Milgrom Model

Suppose that some traders have better information than dealers: assume that the order at time t is placed by a risk neutral *informed* trader with probability π . An informed trader is defined as one with advance knowledge of v . With the complementary probability, $1 - \pi$, the order comes from a *liquidity* trader, who places a market buy or sell order with probability $\frac{1}{2}$ each. A liquidity trader is an investor who trades for reasons unrelated to information about the value of the security: he may be an individual who needs cash for an unanticipated

⁷ For instance, in the extreme case where *all* investors are better informed than the market maker, he will be singled out for a trade *only* when his quotes generate a loss for him. Clearly no one would want to make a market under these circumstances.

⁸ Like Glosten and Milgrom (1985), we suppose that all market participants are risk neutral, market makers are perfectly competitive, and traders place orders of a fixed size. But we simplify their setting by assuming that the value of the security has a binary distribution and that the orders placed by liquidity traders are price inelastic.

contingency, or a fund manager, who has to invest a recent cash inflow or rebalance the portfolio.

For simplicity, the terminal value v has binary distribution, that is, it can take two values: v^H or v^L , with $v^H > v^L$. Let θ_t and $1 - \theta_t$ be the probabilities that dealers assign to the occurrence of a high value, v^H , and a low value, v^L . These probabilities reflect the market makers' views *before* observing the $(t + 1)^{th}$ order. The dealers' estimate of the value *after* the t^{th} order is therefore:

$$\mu_t = \theta_t v^H + (1 - \theta_t) v^L. \quad (3.6)$$

As is explained below, dealers' beliefs about the value of the security evolve because the order flow brings new information. In reality, dealers also get information from other channels (economics news, corporate announcements, etc.). But in order to focus on the informational role of the order flow, here we assume that orders are the sole source of new information for dealers. Formally, this means that $\Omega_t = \{d_1, d_2, \dots, d_t\}$.

Notice that we include the order submitted at date t in the information set used by dealers to determine their quotes at this date.⁹ How can that be, if dealers set their quotes *before* observing the order flow at date t ? Here, it is important to recall that bid and ask quotes are *contingent* offers: a bid price is the price the dealer offers contingent on receiving a sell order ($d_t < 0$), and the ask price is an offer contingent on receiving a buy order ($d_t > 0$). Thus, the answer to the question, "What is the price at which I should execute an order?" must be identical before and after the order is actually received (in this sense, quotes are "with no regrets").

The setting is very similar to the framework developed in Section 3.2, with the difference that now we are more specific about the source of dealers' information. Thus, using the findings of Section 3.2, the ask and bid quotes set by competitive dealers will be:

$$\begin{aligned} a_t &= \mu_t^+ = E(v | \Omega_{t-1}, d_t = +1), \\ b_t &= \mu_t^- = E(v | \Omega_{t-1}, d_t = -1), \end{aligned} \quad (3.7)$$

where μ_t^+ is the dealers' estimate of the security's final value if they receive a buy order at time t , and μ_t^- if they receive a sell order. If the order flow contains information, then it affects dealers' beliefs and we necessarily have $\mu_t^+ \neq \mu_t^-$; hence; $a_t \neq b_t$.

⁹ We also assume that all dealers have identical information, which means in this case that they all observe the order flow realized at each date. This is a reasonable assumption in a market where trade characteristics (price and size) are reported immediately to all market participants, as in many equity markets. But it is unrealistic in more opaque markets, such as the FX market or the bond market. We discuss these points in greater detail in Chapter 8.

How does the order flow affect dealers' value estimate? For instance, on receiving a buy order should they revise it upward or downward? The answer depends on the *correlation* between the order flow and the value, which is ultimately determined by the way informed traders behave. In this model, they are assumed to be risk neutral and to have only a single trading opportunity. They therefore simply maximize their expected gain from trading, $(v - p_t)d_t$, where d_t is their order and p_t is the transaction price (a_t for a buy and b_t for a sell order). As long as bid and ask quotes lie between v^L and v^H (which will be seen to hold in equilibrium), informed investors will always buy ($d_t = +1$) if they observe $v = v^H$ and sell ($d_t = -1$) if they observe $v = v^L$.

This observation has an immediate implication: the likelihood of a buy order is greater when $v = v^H$ than when $v = v^L$. Conversely, the likelihood of a sell order is lower when $v = v^H$ than when $v = v^L$. This is easily checked. A buy order arises in either of two ways: with probability $1 - \pi$ a liquidity trader arrives and buys with probability $1/2$; with probability π , an informed trader arrives and buys (with probability 1) only if $v = v^H$. Hence, if $v = v^H$, the probability of a buy order is $(1 - \pi)/2 + \pi = (1 + \pi)/2$; but if $v = v^L$, the probability is just $(1 - \pi)/2$ because the informed traders do not buy. Thus, whenever $\pi > 0$, the market maker is more likely to receive a buy order when the security has high rather than low value. For sell orders, the converse is true.

This observation captures the intuitive idea that a sequence of buy (sell) orders is more likely when informed traders are privy to good (bad) news about the actual value of the security. Thus, informed trading induces a positive correlation between order flow and value. Buy orders are in fact good news: a *signal* that the true value of the security is high. Conversely, sell orders are bad news, signalling low true value. Accordingly, dealers must mark their value estimate up when they receive a buy order, and down when they get a sell order:

$$E(v | \Omega_{t-1}, d_t = +1) > E(v | \Omega_{t-1}, d_t = -1),$$

which implies, $a_t > b_t$. The bid-ask spread is therefore a natural consequence of the fact that buy and sell orders convey different messages: a dealer is willing to execute buy orders at a premium and sell orders at a discount relative to his own estimate μ_{t-1} of the security's value before the t^{th} transaction.

3.3.2 The Determinants of the Bid-Ask Spread

We now compute the ask and bid prices in this model. First consider the determination of the ask price at time t . In a perfectly competitive market with risk neutral dealers, it will be driven down to the point where dealers make zero

expected profits, that is, the point at which, on average, the losses they make on trading with informed investors are offset by their profits on business with liquidity traders. The market maker must consider that he may be in one of two possible situations:

1. He may sell to an informed investor and so lose $a_t - v^H$, as informed investors buy only when $v = v^H$. At time $t - 1$, when he sets the ask price a_t , the market maker assigns a probability $\pi\theta_{t-1}$ to an informed buy (i.e., the probability π of an informed investor being active on the market multiplied by the probability θ_{t-1} that the asset has value v^H).
2. He may sell to a liquidity trader and so book a profit of $a_t - \mu_{t-1}$: *conditional on trading with a liquidity trader*, the dealer's estimate of the final value of the security remains unchanged at μ_{t-1} since no new information has emerged since time $t - 1$. The probability of receiving a buy order from a liquidity trader is $(1 - \pi)/2$ (i.e., the probability $1 - \pi$ of a liquidity trader being active on the market multiplied by the probability $\frac{1}{2}$ that he buys).

The possible trading events that can occur on the buy and sell sides at time t are summarized in Table 3.2.

Using these probabilities, we can write the dealer's expected net profit from transactions at the ask price a_t :

$$\underbrace{\theta_{t-1}\pi \cdot (a_t - v^H) + 0 \cdot (a_t - v^L)}_{\text{expected profit from trading with informed customer}} + \underbrace{\frac{1}{2}(1 - \pi) \cdot (a_t - \mu_{t-1})}_{\text{expected profit from trading with uninformed customer}} + \underbrace{\left\{(1 - \theta_{t-1})\pi + \frac{1}{2}(1 - \pi)\right\} \cdot 0}_{\text{probability of no ask-side customer}} \cdot 0. \quad (3.8)$$

To give intuition some help, we focus first on the special case of a market maker who is choosing, at time 0, the first ask and bid quotes of the trading day,

Table 3.2 Transaction Probabilities and Underlying Values by Direction of Trade and Trader Identity

Transaction	Identity of trader	Joint probability	Conditional value
Buyer ($d_t = +1$) at a_t	Informed: $v = v^H$	$\theta_{t-1}\pi$	v^H
	Informed: $v = v^L$	0	v^L
	Uninformed	$(1 - \pi)/2$	μ_{t-1}
Seller ($d_t = -1$) at b_t	Informed: $v = v^H$	0	v^H
	Informed: $v = v^L$	$(1 - \theta_{t-1})\pi$	v^L
	Uninformed	$(1 - \pi)/2$	μ_{t-1}

a_1 and b_1 . Assume that this market maker estimates the security's value at its unconditional mean $\mu_0 = (v^H + v^L)/2$ (i.e., assigns probability $\theta_0 = 1/2$ to the value being high). Then the expression for the expected net profit (3.8) becomes:

$$\underbrace{\frac{1}{2}\pi \cdot (a_1 - v^H)}_{\text{expected profit from trading with informed customer}} + \underbrace{\frac{1}{2}(1 - \pi)(a_1 - \mu_0)}_{\text{expected profit from trading with uninformed customer}}. \quad (3.9)$$

If dealers are competitive, a_1 will be such that this expected profit is zero, so that:

$$a_1 = \mu_0 + \pi(v^H - \mu_0) = \mu_0 + \underbrace{\frac{\pi}{2}(v^H - v^L)}_{s_1^a}. \quad (3.10)$$

Hence the ask price at time 1 includes a markup $s_1^a = \frac{\pi}{2}(v^H - v^L)$ over the initial best estimate of the security's value. Naturally, this markup occurs only if there are informed traders ($\pi > 0$) and if they have price-relevant information (that is, $v^H > v^L$). Symmetrically, the bid price b_1 will be:

$$b_1 = \mu_0 + \pi(v^L - \mu_0) = \mu_0 - \underbrace{\frac{\pi}{2}(v^H - v^L)}_{s_1^b}, \quad (3.11)$$

so that the bid price is at a discount $s_1^b = \frac{\pi}{2}(v^H - v^L)$ on the initial estimate. Hence, the bid-ask spread for the first transaction of the day will be:

$$S_1 \equiv a_1 - b_1 = \pi(v^H - v^L). \quad (3.12)$$

This makes it clear that the bid-ask spread is a compensation required by dealers to cover the loss they incur when trading with better informed investors. This loss is often referred to as the cost of adverse selection, and accordingly known as the adverse-selection cost component of the spread. And while it refers only to the first spread of the day, equation (3.12) reveals that the adverse selection cost is an increasing function of two variables:

1. *The proportion of informed traders, π* : The more prevalent are the orders placed by informed traders, the greater the risk of loss for dealers, who accordingly require more compensation for supplying liquidity. This shows how asymmetric information generates illiquidity, motivating

policies, such as bans on insider trading and disclosure requirements, that are intended to reduce informational asymmetries.

2. *The volatility of the security's value, as proxied by its range of variation, $v^H - v^L$:* The broader the range of possible values, $v^H - v^L$, the larger the losses incurred by dealers in trading with informed investors. Accordingly, dealers require more compensation (i.e., a wider bid-ask spread) as volatility increases. And in practice, other things being equal, spreads tend to be wider for more volatile securities (see Stoll 2000, table I).

The same reasoning allows us to derive the quotes the dealer will set later during the day, after revising his value estimate on the basis of the orders received: the probability θ_{t-1} that he places on the security being of high value, at some later time $t - 1$, will presumably no longer be $1/2$. Let us turn back to expression (3.8) for the profit the market maker expects from time t transactions at the ask. Setting this equal to zero, we get the competitive ask price at time t :

$$\begin{aligned} a_t &= \mu_{t-1} + \frac{\pi\theta_{t-1}}{\pi\theta_{t-1} + (1-\pi)\frac{1}{2}} (v^H - \mu_{t-1}) \\ &= \mu_{t-1} + \underbrace{\frac{\pi\theta_{t-1}(1-\theta_{t-1})}{\pi\theta_{t-1} + (1-\pi)\frac{1}{2}}}_{s_t^a} (v^H - v^L). \end{aligned} \quad (3.13)$$

Thus, at any time during the trading day, dealers set their ask price a_t above the fundamental value μ_{t-1} . From the first expression, one sees that the difference $a_t - \mu_{t-1}$ is proportional to the probability of receiving an order from an informed trader, conditional on it being a buy:¹⁰

$$b_t = \mu_{t-1} - \underbrace{\frac{\pi\theta_{t-1}(1-\theta_{t-1})}{\pi(1-\theta_{t-1}) + (1-\pi)\frac{1}{2}}}_{s_t^b} (v^H - v^L), \quad (3.14)$$

¹⁰ Denoting by I the arrival of a buy order from an informed trader, U the arrival of a buy order from an uninformed trader, and B the arrival of a buy order, the dealer's probability of trading with an informed investor on the buy side is:

$$Pr(I|B) = \frac{Pr(I \cap B)}{Pr(B)} = \frac{Pr(I \cap B)}{Pr(I \cap B) + Pr(U \cap B)} = \frac{\pi\theta_{t-1}}{\pi\theta_{t-1} + (1-\pi)\frac{1}{2}}.$$

The second expression is obtained by replacing the term μ_{t-1} in brackets with the lagged value of expression (3.6), that is, rewriting it in terms of the probabilities of the high and low outcomes. The same reasoning yields the bid price b_t (left to the reader as an exercise).

implying that dealers bid to buy the security at a discount s_t^b relative to μ_{t-1} . Using equations (3.13) and (3.14), we obtain the bid-ask spread S_t at time t :

$$S_t \equiv a_t - b_t = s_t^a + s_t^b \quad (3.15)$$

$$= \pi\theta_{t-1}(1 - \theta_{t-1}) \left(\frac{1}{\pi\theta_{t-1} + (1 - \pi)\frac{1}{2}} + \frac{1}{\pi(1 - \theta_{t-1}) + (1 - \pi)\frac{1}{2}} \right) (v^H - v^L).$$

This expression for the spread is more general than (3.12), which is the special case of $\theta_{t-1} = \frac{1}{2}$. It remains true that the spread increases with the proportion of informed traders (π) and the volatility of the security's value ($v^H - v^L$). In addition, expression (3.15) shows that the adverse selection trading cost S_t has a third determinant, namely, dealers' beliefs about the value of the security, θ_{t-1} . Unlike the first two, this determinant varies over time, as the market maker changes his value estimate. The spread is greatest when $\theta_{t-1} = 0.5$, as posited in expression (3.12), and falls to zero as θ_{t-1} goes to 1 or 0. To understand this result, suppose that θ_{t-1} is close to 1. This means that dealers are quite confident that the fair value of the security is v^H , so a new buy order is not a surprise—no news—and does not move their value estimate: a_t is very close to μ_{t-1} . But when $\theta_{t-1} = 0.5$, dealers are highly uncertain about the direction of the market, so their beliefs are highly sensitive to the new orders they receive. Accordingly, the revision of their value estimates is substantial. Therefore the spread is greater when dealers are more uncertain about the value of the security. This is why bid-ask spreads often increase in advance of the release of important information, such as macroeconomic data or earnings announcements. This may also explain why the spreads tend to be larger at market openings.

Let us highlight two features of this model. First, it is a highly stylized model of market making, with adverse selection costs the sole component of the bid-ask spread. As we shall see in Sections 3.4 and 3.5, there may be additional determinants as well. Second, since in this model dealers' expected profit is zero, the adverse selection cost eventually bears on the liquidity traders, who—by “paying” the bid-ask spread—lose exactly what informed traders gain. In our setting, liquidity traders' demand is inelastic: it is independent of the price. Intuitively, in reality, one would expect them to be less likely to place orders when the bid-ask spread is greater. As exercise 6 shows, this situation can lead to a market breakdown, where the spread is so wide that no trade occurs at all. This suggests that asymmetric information may also lead to a complete market freeze, as in the interbank market during the crisis of 2007–8.¹¹

¹¹ The insight that adverse selection can shut a market down dates back to Akerlof's celebrated piece in 1970 on the market for “lemons.”

3.3.3 How Do Dealers Revise their Quotes?

Up to now, in deriving the zero-profit bid and ask prices that competitive dealers will set, we have taken their beliefs about the underlying value of the security as *given*, that is, μ_{t-1} and θ_{t-1} are given. As bid-ask quotes depend on these beliefs (see equation (3.15)), in order to understand how quotes evolve over time, we must explain how dealers form their beliefs in the light of new orders. For instance, on receiving a flurry of buy orders, a dealer should attach more and more weight to the possibility that the security has a high value.

To see how this is done, consider how dealers receiving a buy or sell order at time t , revise the probability that they attach to the value being high, starting from an initial probability θ_{t-1} . Define θ_t^+ as the probability that dealers assign to high value, $v = v^H$, after they receive a buy order at date t , and let θ_t^- be the corresponding probability in the wake of a sell order. Formally,

$$\begin{aligned}\theta_t^+ &\equiv \Pr(v = v^H \mid \Omega_{t-1}, d_t = +1), \\ \theta_t^- &\equiv \Pr(v = v^H \mid \Omega_{t-1}, d_t = -1).\end{aligned}$$

To compute θ_t^+ using Bayes' Rule,¹² let A be the event $v = v^H$ and B the arrival of a buy order at time t . Then $\Pr(A) = \theta_{t-1}$; $\Pr(B) = \pi\theta_{t-1} + (1-\pi)\frac{1}{2}$ (from Table 3.2); and the probability of a buy order when $v = v^H$ is $\Pr(B|A) = \pi \cdot 1 + (1-\pi)\frac{1}{2} = (1+\pi)\frac{1}{2}$, since in this case any informed trader will be a buyer. So the probability that $v = v^H$, conditional on a buy order at time t , is given by:

$$\theta_t^+ = \Pr(A|B) = \frac{\Pr(B|A) \cdot \Pr(A)}{\Pr(B)} = \frac{(1+\pi)\frac{1}{2} \cdot \theta_{t-1}}{\pi\theta_{t-1} + (1-\pi)\frac{1}{2}}.$$

The derivation of θ_t^- is analogous:

$$\theta_t^+ = \frac{(1+\pi)\frac{1}{2}}{\pi\theta_{t-1} + (1-\pi)\frac{1}{2}}\theta_{t-1}, \quad (3.16)$$

$$\theta_t^- = \frac{(1-\pi)\frac{1}{2}}{\pi(1-\theta_{t-1}) + (1-\pi)\frac{1}{2}}\theta_{t-1}. \quad (3.17)$$

¹² Bayes's rule states that the conditional probability of an event A given an event B is:

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)} = \frac{\Pr(B|A) \cdot \Pr(A)}{\Pr(B)}.$$

It is easy to see that $\theta_t^+ > \theta_{t-1}$. Dealers become more bullish about the security when they get a buy order, insofar as buy orders signal that informed traders may have good news. In a symmetrical way, $\theta_t^- < \theta_{t-1}$: dealers become more bearish when they get a sell order, as this indicates that informed traders may have bad news.

Upon receiving a buy order at time t , the dealers' updated expectation of the security's value is the weighted average of v^H and v^L , where θ_t^+ and $1 - \theta_t^+$ are the updated probability weights:

$$\mu_t^+ = \theta_t^+ v^H + (1 - \theta_t^+) v^L. \quad (3.18)$$

Recalling that the dealers' estimate before receiving the buy order was μ_{t-1} , their revision of the value in the wake of a buy order is:

$$\begin{aligned} \mu_t^+ - \mu_{t-1} &= \theta_t^+ v^H + (1 - \theta_t^+) v^L - [\theta_{t-1} v^H + (1 - \theta_{t-1}) v^L] \\ &= \frac{\pi \theta_{t-1} (1 - \theta_{t-1})}{\pi \theta_{t-1} + (1 - \pi) \frac{1}{2}} (v^H - v^L) = s_t^a. \end{aligned} \quad (3.19)$$

Analogously, in the wake of a sell order:

$$\begin{aligned} \mu_t^- - \mu_{t-1} &= \theta_t^- v^H + (1 - \theta_t^-) v^L - [\theta_{t-1} v^H + (1 - \theta_{t-1}) v^L] \\ &= \frac{\pi \theta_{t-1} (1 - \theta_{t-1})}{\pi (1 - \theta_{t-1}) + (1 - \pi) \frac{1}{2}} (v^H - v^L) = s_t^b. \end{aligned} \quad (3.20)$$

So s_t^a and s_t^b can be interpreted as the changes to dealers' estimates of the asset value: they correspond to ε_{t+1} in the model described in Section 3.2. The model here simply recognizes that the order flow itself is a source of information when orders can come from informed traders. Now that we know the dynamics of dealers' beliefs, we also know how they will set their quotes. As equation (3.7) shows, they will set their ask price in anticipation of a buy and therefore will set $a_t = \mu_t^+$. Conversely, they will bid $b_t = \mu_t^-$.¹³

¹³ Note that the ask price at time t can be expressed in terms of the belief θ_{t-1} only (without reference to μ_{t-1}) by using (3.16) in (3.18), and similarly for the bid price:

$$\begin{aligned} a_t = \mu_t^+ &= \frac{(1 + \pi) \theta_{t-1}}{2\pi \theta_{t-1} + 1 - \pi} v^H + \frac{(1 - \pi)(1 - \theta_{t-1})}{2\pi \theta_{t-1} + 1 - \pi} v^L \\ b_t = \mu_t^- &= \frac{(1 - \pi) \theta_{t-1}}{2\pi (1 - \theta_{t-1}) + 1 - \pi} v^H + \frac{(1 + \pi)(1 - \theta_{t-1})}{2\pi (1 - \theta_{t-1}) + 1 - \pi} v^L. \end{aligned}$$

Note that we have used two different but equivalent ways to interpret and derive bid and ask prices. In Section 3.3.2, we obtained them by positing that dealers set price markups s_t^a and s_t^b so as to offset the potential losses on trades with better informed agents with the profits on business with uninformed traders. Here, instead, as illustrated by equations (3.19) and (3.20), we derived the markups s_t^a and s_t^b as the revisions to the dealers' estimate of the value of the security following buy and sell orders. That this change is equal to the markup required by dealers to execute a buy order reflects the fact that the order flow itself contains information.

To sum up, the wedge between the dealers' ask price and their prior estimate of fundamentals is both a compensation for the risk of trading with better informed investors *and* an adjustment to the information contained in order flow. Of course, the two approaches produce the same result: equations (3.19) and (3.20) yield the same ask and bid prices as equations (3.13) and (3.14). The advantage of looking at the revision in beliefs is that it enables us to characterize how dealers change their quotes over time in response to the order flow. Therefore, if we specify a time series for the order flow, we can now map out the corresponding time path of quotes, as we do in the next section.

3.3.4 Price Discovery

Buy orders execute at the ask price, sell orders at the bid, so the transaction price at date t is:

$$P_t = \begin{cases} a_t = \mu_t^+ & \text{if } d_t = +1, \\ b_t = \mu_t^- & \text{if } d_t = -1. \end{cases} \quad (3.21)$$

Hence:

$$p_t = \mu_t = \theta_t v^H + (1 - \theta_t) v^L, \quad (3.22)$$

where $\theta_t = \theta_t^+$ if $d_t = +1$, and $\theta_t = \theta_t^-$ if $d_t = -1$. Thus, transaction prices reflect all the information available to market makers at time t , which is to say the information Ω_{t-1} they had at time $t-1$ plus the direction of the order flow that they receive at time t .

Can we therefore conclude that transaction prices always reflect all available information, as claimed by the EMH? The notion of "all available information" is problematic in an environment in which traders have asymmetric information. Does it mean both public and private information, or only public information?

In other words, in this context does the EMH apply in strong, semi-strong, or weak form?

The semi-strong form states that transactions take place at fair values given all *public* information. In our model, public information at date t is the order flow observed up to this date, $\Omega_t = \{d_1, d_2, \dots, d_t\}$. As $p_t = \mu_t = E(v | \Omega_t)$, the semi-strong EMH holds true in this model. This might seem surprising, insofar as the bid-ask spread is usually construed as a friction that causes temporary deviations of transaction prices from fundamental asset values. But in this model the difference between the execution prices for buy orders and sell orders is due entirely to the fact that they convey different information. Thus, far from being a symptom of inefficiency, the bid-ask spread is part of the mechanism by which dealers incorporate the information contained in the order flow into the price process.

The strong version of the EMH states that transactions always take place at fair values given all public *and* private information. For instance, suppose that informed traders learn that $v = v^H$. If the EMH holds in strong form, then all trades take place at prices equal to v^H (insofar as market participants get no further information). At first glance, it seems impossible for markets to be informationally efficient in this sense. How could prices ever reflect information that is not yet available to all market participants? Yet in this model this is a possibility. To see this, suppose that $\pi = 1$. In this case, the quotes posted by dealers at time 1 (setting $\pi = 1$ in equations (3.13) and (3.14)) will be:

$$\begin{aligned} a_1 &= v^H, \\ b_1 &= v^L. \end{aligned}$$

If $v = v^H$, then informed traders buy and the first transaction of the day is crossed at $p_1 = a_1 = v^H$.¹⁴ Upon observing this transaction, dealers infer that the value of the security is high since all traders are informed and they are buying. Thus, $\theta_1 = 1$, and all subsequent transactions are at v^H (only insofar as no further information arrives, of course). Similar reasoning applies when $v = v^L$. Thus, the strong form version of the EMH holds, despite the fact that dealers initially have less information than some market participants. This is because dealers draw rational inferences from the order flow; that is, they *learn* about private information from the order flow and adjust their quotes accordingly.

Of course, this example is extreme (positing that $\pi = 1$), but it does suggest that prices can adjust gradually to the fair value of the security as dealers learn it

¹⁴ In this case, informed investors are just indifferent between buying or not trading. We assume that they buy.

from the order flow. The process by which private information is incorporated into prices and the market becomes informationally efficient is called *price discovery*. Thus, one advantage of the model presented here is that it allows us to study whether and how fast price discovery occurs.

Ultimately these questions bear on the dynamics of dealers' beliefs about the asset's fair value. For instance, suppose as before that $v = v^H$. Equation (3.22) shows that transaction prices will converge to v^H if θ_t goes to 1, in other words, if dealers are increasingly confident that the value of the security is high. Thus, the speed of price discovery is the speed at which θ_t goes to 1. In this model, it can be shown formally that price discovery always obtains in the long run if there are some informed traders in the market ($\pi > 0$) and that its speed increases with the fraction π of informed investors.

A formal derivation of these claims is tedious, but the intuition is simple. In the absence of informed trading, the order flow is balanced, with 50% buy and 50% sell orders. However, with informed trading, the flow is unbalanced and its tendency provides information on value. For instance, if $v = v^H$, the proportion of buy orders will be more than 50%; if $v = v^L$, sell orders will exceed 50%. The speed of the learning process is then determined by how fast dealers become confident that buy orders make up more or less than 50%, which in turn is determined by the proportion of informed traders: buy orders will dominate very quickly if π is close to 1 and $v = v^H$; sell orders will predominate if $v = v^L$.

A good way to illustrate these intuitions is to simulate the model. Assume that $v^H = 102$, $v^L = 98$, and $\theta_0 = 0.5$. Thus, at the start of the trading day, dealers estimate the asset value at $\mu_0 = 100$. Initially, suppose that $\pi = 0.3$ and that the true value of the security is $v = v^H$. On this basis, we draw a sequence of one hundred orders (each of which is a buy with probability $\pi + (1 - \pi)/2 = 0.65$), and record the evolution of dealers' beliefs θ_t associated with this specific sequence of order arrivals, as well as the evolution of transaction prices p_t . For instance, suppose that the two first orders are buys and the third is a sell. Equation (3.16) yields the corresponding dealers' beliefs: $\theta_1 = 0.65$, $\theta_2 = 0.77$, and $\theta_3 = 0.65$.¹⁵ Equation (3.22) gives the corresponding sequence of transaction prices: $p_1 = 100.6$, $p_2 = 101.10$, $p_3 = 100.6$.

¹⁵ A convenient way to express the updating of dealers' beliefs θ_t is to recast equations (3.16) and (3.17) in terms of the odds ratio, which yields a linear first-order difference equation:

$$\frac{\theta_t^+}{1 - \theta_t^+} = \frac{1 + \pi}{1 - \pi} \cdot \frac{\theta_{t-1}}{1 - \theta_{t-1}} \quad \text{and} \quad \frac{\theta_t^-}{1 - \theta_t^-} = \frac{1 - \pi}{1 + \pi} \cdot \frac{\theta_{t-1}}{1 - \theta_{t-1}}.$$

Therefore the odds ratio at time t is simply a function of the order imbalance x_t , defined as the cumulative difference between buy and sell orders up to time t :

$$\frac{\theta_t}{1 - \theta_t} = \frac{\theta_0}{1 - \theta_0} \left(\frac{1 + \pi}{1 - \pi} \right)^{x_t}, \text{ where } x_t \equiv \sum_{\tau=1}^t d_\tau.$$

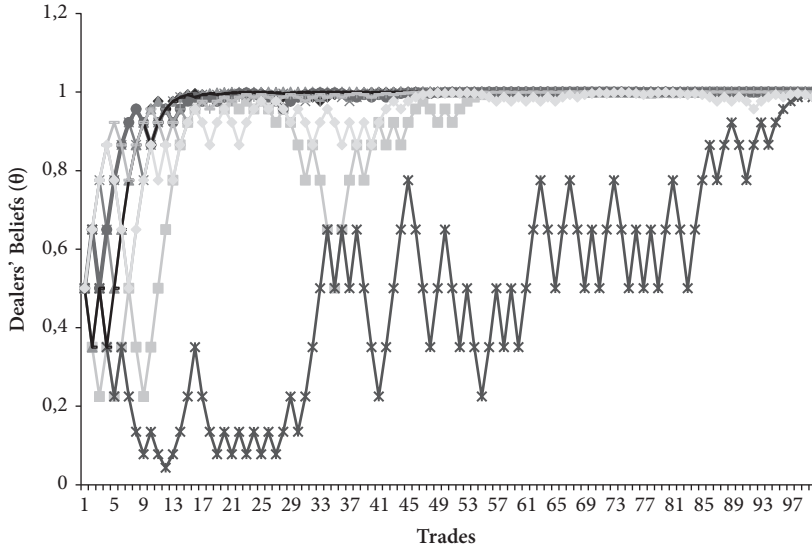


Figure 3.1 Evolution of dealers' beliefs.

We repeat this experiment ten times, drawing ten different random sequences of one hundred orders with the same value of π . Figure 3.1 shows the pattern of dealers' beliefs in each experiment. Clearly, the probability assigned to the high value by the dealers converges to 1 (they eventually discover the correct value), and relatively quickly in most cases.

Next, we measure price discovery by computing the squared difference between the transaction price p_t at date t and v^H ("the pricing error"):

$$PD_t = (p_t - v^H)^2. \quad (3.23)$$

As dealers learn about the value of the security, PD_t goes to zero; that is, it is an inverse measure of price discovery. Figure 3.2 shows the corresponding evolution for the pricing error in each experiment. Again, in most cases it declines quickly as dealers discover the true value.

To see how the proportion of informed traders π affects price discovery, we run the experiment again with three different values of π : 0.1, 0.5, and 0.9. Figure 3.3 shows the evolution of the squared pricing errors, PD_t (for each value of π) averaged across the ten simulations: $\sum_{i=1}^{i=10} PD_{it}/10$. The middle line summarizes the results for $\pi = 0.5$: on average, full price discovery is achieved after fifteen trades. The bottom line shows that when informed trading is predominant, to the point that nine in ten orders are placed by informed traders, price discovery speeds up considerably: on average, full price discovery

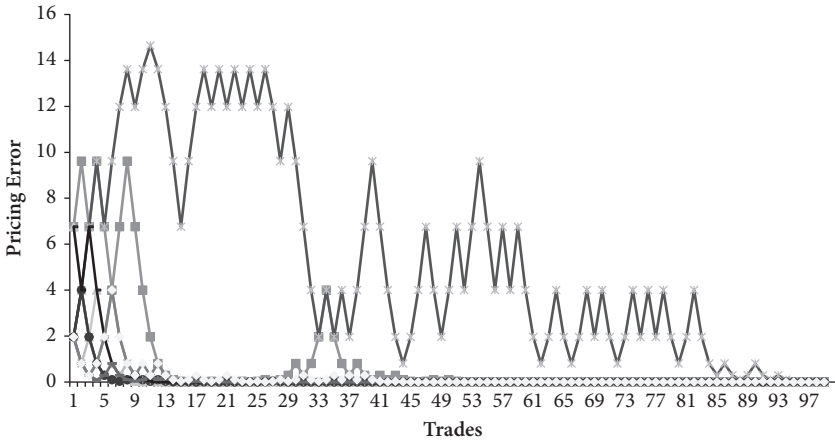


Figure 3.2 Evolution of pricing errors.

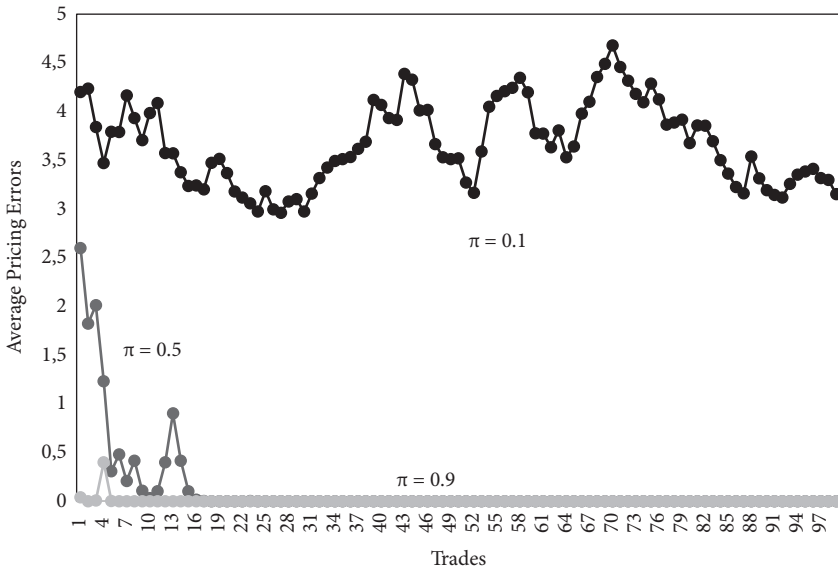


Figure 3.3 Speed of price discovery and the level of informed trading.

occurs after just five trades. In contrast, when informed trading is rare (only one in ten orders placed by informed traders), convergence is much slower—almost imperceptible even after one hundred transactions—because the signal-to-noise ratio in the order flow is much smaller.

These observations on price discovery pose a conundrum for market organizers: more frequent informed trading widens bid-ask spreads and tends to make

the market illiquid, as shown in equation (3.15), but it also speeds price discovery. Intuitively, the presence of informed traders is necessary for information to be incorporated in prices. At least in the short run, therefore, there is a trade off between liquidity and informational efficiency.

3.3.5 The Implications for Price Movements and Volatility

Now let us go back to the trade and quote data for the AGF stock in the introduction (Table 3.1). A graphical representation of the price dynamics for AGF is given in Figure 3.4 (considering the first fifty trades on March 26, 2001).

The upper and lower curves chart the best ask and bid quotes, respectively. The dots indicate the prices at which trades take place. It is apparent from the figure that prices adjust in the direction of the order flow. For instance, most of the first ten trades are initiated by sellers placing market orders, and the midprice declines. Conversely, trades twenty-five to forty are mostly triggered by buy market orders, and the midprice rises.

The model developed in this section provides a framework for explaining this effect of order flow on price movements. Recall that (see equations (3.19) and (3.20)):

$$\begin{aligned}\mu_t^+ &= \mu_{t-1} + s_t^a, \\ \mu_t^- &= \mu_{t-1} - s_t^b,\end{aligned}$$

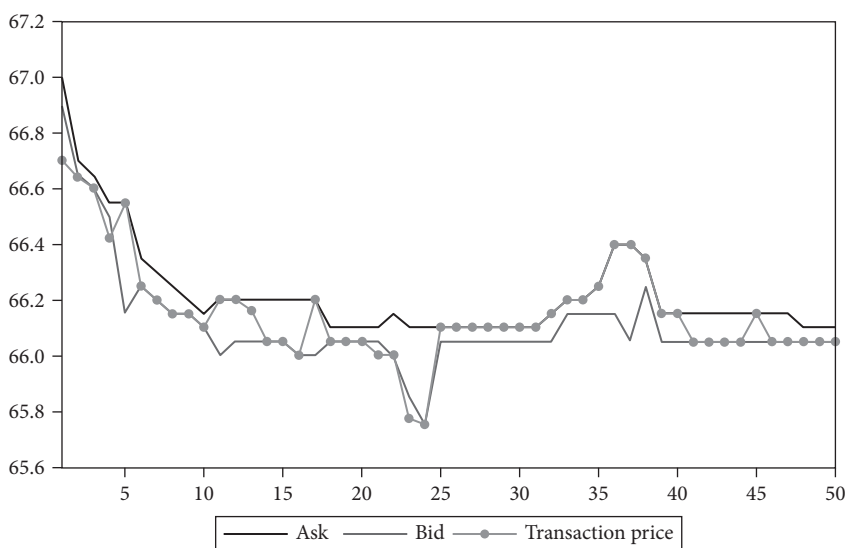


Figure 3.4 Quotes and transaction prices for AGF (fifty first trades of the day).

where the expressions for s_t^a and s_t^b are given by equations (3.13) and (3.14). These equations can be written more compactly as:

$$\mu_t = \mu_{t-1} + s(d_t) d_t, \quad (3.24)$$

where:

$$s(d_t) = \begin{cases} s_t^a & \text{if } d_t = +1 \\ s_t^b & \text{if } d_t = -1. \end{cases}$$

The dealers' estimate of the value of the security after the t^{th} transaction, μ_t , depends on the direction of the order flow; market orders are informative.

Now recall that dealers set their quotes at time t so as to bracket their prior estimate of the security's value, μ_{t-1} , since:

$$a_t = \mu_t^+ = \mu_{t-1} + s_t^a, \quad (3.25)$$

$$b_t = \mu_t^- = \mu_{t-1} - s_t^b. \quad (3.26)$$

Thus they will revise their quotes upward after buy orders and downward after sell orders, exactly as observed in Figure 3.4 for AGE. As an illustration, consider the numerical example analyzed in the previous section ($v^H = 102$, $v^L = 98$, $\theta_0 = \frac{1}{2}$, and $\pi = \frac{1}{2}$). On these assumptions we readily find, from equations (3.13) and (3.14), that $a_1 = 101$ and $b_1 = 99$. Now, assume that the first order is a buy ($d_1 = +1$). This leads dealers to revise the probability of $v = v^H$ upward to $\theta_1 = \frac{3}{4}$ and correspondingly mark up their value estimate to $\mu_1 = 101$. Again using equations (3.13) and (3.14), dealers will raise their bid and ask quotes to $a_2 = 101.6$ and $b_2 = 100$ respectively. If the first order is a sell order, then $\mu_1 = 99$ instead, and $\theta_1 = \frac{1}{4}$. After this trade, dealers lower their bid and ask quotes to $a_2 = 100$ and $b_2 = 98.4$.

As a result, trade-to-trade changes in transaction prices are correlated with the order flow. All transactions take place either at the ask price or the bid price, so that $p_t = \mu_t$. Therefore, equation (3.24) also implies:

$$p_t - p_{t-1} = \mu_t - \mu_{t-1} = s(d_t) d_t. \quad (3.27)$$

Hence, in this framework the difference between the price of the $(t-1)^{\text{th}}$ transaction and the t^{th} transaction is entirely determined by the direction of the t^{th} order.¹⁶ The last equation is similar to equation (3.4), except that it explicitly

¹⁶ Equation (3.27) implies that the price change in any given time period is proportional to the order imbalance during the period. Thus, it lays the foundation for the price impact regression discussed in Section 2.3.2. Here, the order size is fixed, so only the direction of orders is informative. Chapter 4 considers a more complex environment in which both order direction and size are informative.

relates the change in price ε_t to the order flow d_t : here the price innovation ε_t is equal to $s(d_t)d_t$. This makes sense because, under asymmetric information, the order flow is informative.

Finally, from equation (3.27) we can compute the variance of price changes:

$$\text{var}(\Delta p_t) = \text{var}(s(d_t)d_t). \quad (3.28)$$

Return volatility at time t is thus determined both by the size of the bid-ask spread $s(d_t)$ and by the uncertainty of the direction of the order flow d_t . This shows that trading is also necessarily a source of volatility, being a source of information. Moreover, equation (3.28) suggests that return volatility might not be constant during the trading day, and that it should be correlated with the bid-ask spread. As dealers' value estimate becomes more accurate, their quotes should become less sensitive to order flow, the spread should narrow, and return volatility should decrease. Interestingly, this pattern is observed in equity markets: from the opening to the mid-session, both volatility and bid-ask spreads decline on average, before picking up again as the market close approaches. Of course, our interpretation presupposes that private information is obtained mostly at the start of the trading day or before: if during the day dealers come to suspect that some new price-relevant information—say, a rumor of a takeover bid—is guiding the order flow, they will widen their spreads again, to protect themselves against traders who may have a better sense of the likely outcome.

Expression (3.28) for return volatility is based on the idea that new information affects stock prices only via the orders placed by informed traders. In reality, much new information takes the form of public news, which leads all market participants to revise their quotes and therefore leads to price changes without triggering any trades. Hence, the volatility of stock returns also reflects the variance of price changes induced by such public news. Models with informed trading can easily be adapted to include public news, as will be seen in Section 5.2.1.

3.4 Price Dynamics with Order-Processing Costs

The risk of losing money in trading with better informed investors is the only cost borne by liquidity suppliers that we have considered so far. In reality, however, processing a transaction costs time and money: trading fees, clearing and settlement fees, paperwork and back office work, telephone time, and so

on.¹⁷ Naturally, liquidity suppliers also require a compensation for these so-called “order-processing costs.” To explore how they affect the bid-ask spread and the resulting dynamics of transaction prices, we extend the framework developed in the previous section by assuming an order-processing cost equal to γ per share. In reality, some of these costs are per transaction, others per dollar traded or per share traded. Most of our results here would also hold if order-processing costs were γ per dollar traded or per transaction (see exercise 1).

3.4.1 Bid-Ask Spread with Order-Processing Costs

Consider again the determination of the bid-ask spread for, say, the t^{th} transaction. Suppose that a buy order is received. To break even, a dealer’s ask price must now cover not only the expected loss from trading with potentially informed buyers s_t^a but also the order-processing cost γ . So equation (3.25) must be modified as follows:

$$a_t = \mu_{t-1} + \gamma + s_t^a. \quad (3.29)$$

Dealers simply pass the processing cost on to liquidity demanders. By a similar reasoning, the bid price posted by dealers at time t is:

$$b_t = \mu_{t-1} - \gamma - s_t^b. \quad (3.30)$$

Hence, the bid-ask spread:

$$S_t \equiv a_t - b_t = 2\gamma + s_t^a + s_t^b \quad (3.31)$$

now has two different components: order-processing cost (2γ) and adverse-selection cost ($s_t^a + s_t^b$). For practical purposes, it is important to gauge their relative importance in determining the spread. For instance, the policy measures needed to alleviate illiquidity are not the same in the two cases: technological upgrades and rules encouraging competition among trading platforms can reduce processing costs, while action against insider trading may mitigate adverse selection.

How the two components of the bid-ask spread can be measured separately is not obvious, since at first glance they have the same effect. In the next section,

¹⁷ Automation reduced some of these costs considerably, but they are still not negligible.

however, we will see that the two components carry very different implications for the dynamics of transaction prices, which gives us a way to estimate the components of the spread using data on order flow and transaction prices.

3.4.2 Price Dynamics with Order-Processing and Adverse-Selection Costs

As trades take place at bid and ask prices, the transaction price at time t can be written as:

$$p_t = \mu_{t-1} + (s(d_t) + \gamma)d_t, \quad (3.32)$$

where $s(d_t) = s_t^a$ if $d_t = 1$, and $s(d_t) = s_t^b$ if $d_t = -1$. As $\mu_t = \mu_{t-1} + s(d_t)d_t$, we obtain:

$$p_t = \mu_t + \gamma d_t. \quad (3.33)$$

Thus, in the presence of order-processing costs, transaction prices deviate from fair value given all public information, including the direction of trade initiation.

The deviation is equal to the processing cost (as $|p_t - \mu_t| = \gamma$). Intuitively, dealers cover this cost by executing buy (sell) orders at a markup (discount) relative to their value estimate based on all public information.

Thus, order-processing costs induce *transient* deviations of transaction prices from fundamental values, so called because these deviations do not correspond to a revision in dealers' value expectations. Accordingly, they are subsequently corrected, at least in part by opposite price movements ("reversals").

To see this, consider the arrival of a buy order at time t and define its short-term (ST) price impact as the deviation of the transaction price from the immediately preceding fundamental value, μ_{t-1} :

$$\text{ST impact} \equiv p_t - \mu_{t-1} = a_t - \mu_{t-1}.$$

Using equation (3.29), one can see that the immediate effect of a buy market order is to push the price up by:

$$\text{ST impact} = s_t^a + \gamma > 0. \quad (3.34)$$

Now consider the long-term (LT) price impact, that is, its effect on the average price at a distant time $t + T$, p_{t+T} . At that time, by equation (3.33), the price will be given by $p_{t+T} = \mu_{t+T} + \gamma d_{t+T}$. Its expected value as of time t is then:

$$\begin{aligned}
E(p_{t+T} | \Omega_{t-1}, d_t = 1) &= E(\mu_{t+T} | \Omega_{t-1}, d_t = 1) + \gamma E(d_{t+T} | \Omega_{t-1}, d_t = 1) \quad (3.35) \\
&= \mu_t + \gamma E(d_{t+T} | \Omega_{t-1}, d_t = 1) \\
&= \mu_{t-1} + s_t^a + \gamma E(d_{t+T} | \Omega_{t-1}, d_t = 1),
\end{aligned}$$

where the first step follows by the law of iterated expectations and the second by (3.32).

Suppose that $t + T$ is so far ahead that by then the currently pending uncertainty about true value of the asset (v^H or v^L) will have already been resolved—the fundamental publicly revealed (say, $t + T$ is beyond the end of the trading day). Then, the direction of trade at time t (d_t) has no predictive power for its direction at $t + T$ (d_{t+T}). Hence $E(d_{t+T} | \Omega_{t-1}, d_t = 1) = 0$, so that:

$$E(p_{t+T} | \Omega_{t-1}, d_t = 1) = \mu_{t-1} + s_t^a.$$

Thus the long run impact of the buy order at time t includes only the informational component of the bid-ask spread, and not the order-processing cost γ .¹⁸

$$\text{LT impact} = s_t^a. \quad (3.36)$$

Hence, in the long term, only the informational effect persists:

$$\text{ST impact} - \text{LT impact} = \gamma. \quad (3.37)$$

In the absence of order-processing costs ($\gamma = 0$), the short-term impact of the buy order is equal to the long-term impact, because the impact is due entirely to the informational content of the buy order. In contrast, in presence of order-processing costs, the short run impact exceeds the long run impact by an amount equal to the processing cost, creating a further temporary price blip at time t .

¹⁸ Working out the medium-term effect of the buy order on price is somewhat trickier. Suppose that time $t + T$ is not so distant, and that by then the true value of the security is not yet publicly revealed. In this case d_t has some predictive value for the future order flow d_{t+T} . To see this, consider that a buy order at t leads to an upward revision from θ_{t-1} to θ_t^* in the probability that the stock has a high value, as described in equation (3.16), and so it increases the probability of further buy orders from informed investors at $t + T$, that is:

$$E(d_{t+T} | \Omega_{t-1}, d_t = 1) > E(d_{t+T} | \Omega_{t-1}).$$

This increased probability is associated with an additional medium-term price impact beyond s_t^a . To see this, compare the expected price at $t + T$ before and after the buy order at time t :

$$\begin{aligned}
&E(p_{t+T} | \Omega_{t-1}, d_t = 1) - E(p_{t+T} | \Omega_{t-1}) \\
&= s_t^a + \gamma[E(d_{t+T} | \Omega_{t-1}, d_t = 1) - E(d_{t+T} | \Omega_{t-1})] > s_t^a,
\end{aligned}$$

where we have used equation (3.35). In exercise 10 the reader is asked the precise value of the revision in the expectation of d_{t+T} in this expression.

Thus, after the initial price increase, the price level tends to revert. A similar decomposition can be obtained if the order at time t is a sell order: its immediate effect is to reduce the price, but in the long run the price bounces back and partly reverts to its initial level. Thus, with order-processing costs, trade-to-trade changes in prices will show a negative serial correlation. As Chapter 2 explains, Roll's model exploits this feature to derive an estimator of the bid-ask spread and corresponds to a special version of the current model where the bid-ask spread is due only to order-processing costs. These effects can also be observed in Figure 3.4. For instance, the sequence of buy orders from the twenty-fifth to the fortieth transaction trigger a price increase, but part of it is transient so that after the fortieth transaction the price partly reverts.

To summarize, buy orders and sell orders create upward and downward price pressures, which are permanent only to the extent that illiquidity is due to asymmetric information. The transient component is instead a compensation for the costs borne by liquidity suppliers to accommodate temporary order imbalances.

As order-processing costs and asymmetric information typically coexist, market orders have both permanent and temporary effects. This observation can be used to measure order-processing and adverse selection costs, as Chapter 5 explains.

Box 3.1 Dealers' Rents as an Additional Component of the Bid-Ask Spread

In this section, we interpret γ as a cost per share borne by dealers to execute market orders. But γ may equally well be viewed as a measure of dealers' rents from market power. To see this, suppose that order-processing costs are equal to γ^c and that dealers require an expected profit per share *at least* equal to γ^r on each transaction. It follows immediately that to achieve an expected profit of γ^r for executing a buy market order, dealers must charge an ask price equal to:

$$a_t = \mu_{t-1} + \gamma^c + \gamma^r + s_t^a,$$

so that the term $2\gamma = 2(\gamma^c + \gamma^r)$ in expression (3.31) for the bid-ask spread S_t will reflect not only the dealers' operating costs γ^c but also their non-competitive rents γ^r .

By the same token, we can repeat the reasoning of Section 3.4.2 above to conclude that after the execution of a buy market order the price level will revert by an amount averaging $\gamma^c + \gamma^r$. Similar remarks apply to sell orders. Thus, considering only the dynamics of prices after a trade, one cannot

measure separately the order-processing component and the rent component of the bid-ask spread. This implies that what is termed the “order-processing cost” component must be interpreted with care, since it may in fact capture both processing costs proper and the rents captured by dealers in supplying liquidity.

In reality, dealers often earn rents. Indeed, some markets charge a fee on market participants to become dealers. The level of this fee is an indicator of the economic value of being a dealer and is a determinant of the number of dealers. Competition among dealers may be imperfect also due to collusion. In the 1990s, Nasdaq dealers were accused of implicitly colluding on excessive spreads (see Christie, Harris, and Schultz 1994; Christie and Schultz 1994). Chapter 4 offers a more systematic analysis of the way trading rules can affect liquidity suppliers’ ability to earn rents at the expense of liquidity demanders.

3.5 Price Dynamics with Inventory Risk

In continuous markets, buy and sell orders from traders seeking liquidity do not arrive at the same time. This creates temporary order imbalances, and one important function of liquidity suppliers is to serve as counterparty when the order flow is unbalanced. For instance, a dealer may execute a sell order for five thousand shares of a stock, which increases his position in it. Next, ten minutes later, he receives a buy order for one thousand shares, which allows him to pare his extra inventory down. Finally, he receives another buy order for four thousand shares, and so reverts to his initial position. Over time, the net position taken by the dealer is zero. His role was simply to balance demand and supply over time.

However, this role exposes the dealer to inventory risk, that is, the possibility of a change in the value of his holdings because of, say, news about the underlying fundamentals. Assuming that dealers are risk averse, they will require a compensating risk premium, known as inventory holding cost—even if they are not at risk of being picked off by traders with superior information. Thus, inventory risk is another important determinant of the bid-ask spread, as was first shown formally by Stoll (1978). Here we analyze the effect of inventory risk on dealers’ pricing policy and discuss its implications for the spread and for price dynamics.

In order to isolate the impact of inventory holding costs, we make two simplifying assumptions throughout this section. First, the order flow is not correlated with news about fundamentals; that is, it is not driven by traders with an informational advantage over market makers ($\pi = 0$). Second, dealers have no order-processing costs ($\gamma = 0$).

Before trading at any time t , the representative dealer has cash c_t and a starting inventory of the risky security z_t , where $z_t > 0$ indicates a long and $z_t < 0$ a short position.¹⁹ Marked to market, his initial wealth w_t is the value of his cash and security position before trading at time t , evaluated at time t market price p_t :

$$w_t = p_t z_t + c_t. \quad (3.38)$$

The dealer's concern is how his trade at time t will affect his next-period wealth $w_{t+1} = p_{t+1} z_{t+1} + c_{t+1}$, where z_{t+1} and c_{t+1} are his stock inventory and cash position after trading at time t .

The market is organized as a call auction where the representative dealer is assumed to behave competitively, that is, to take the price p_t as given in choosing the number of shares y_t that he is willing to supply:²⁰ $y_t > 0$ indicates the dealer's offer to sell y_t shares, and $y_t < 0$ means that he is willing to bid to buy $|y_t|$ shares. Thus the dealer's inventory after selling y_t shares is:

$$z_{t+1} = z_t - y_t, \quad (3.39)$$

and his corresponding cash position after receiving payment for those shares of $p_t y_t$ is:

$$c_{t+1} = c_t + p_t y_t. \quad (3.40)$$

Thus, the dealer's end-of-period wealth, as a function of the amount y_t he sells at time t , is:

$$w_{t+1} = p_{t+1} z_{t+1} + c_{t+1} = p_{t+1} (z_t - y_t) + c_t + p_t y_t, \quad (3.41)$$

using (3.39) and (3.40) in the definition of w_{t+1} .

For ease of exposition, in Section 3.5.1 we start with a static setting with a single trading period, where dealers trading at time 0 expect to liquidate their holdings at a price equal to the fundamental value at time 1, that is, $p_1 = v = \mu_0 + \varepsilon_1$. Then, in Section 3.5.2, we turn to a simple setting with two trading periods, where initially each dealer anticipates that the subsequent value of his inventory will be determined by the future order flow. In this dynamic setting,

¹⁹ Since we are considering a representative dealer, z_t must be interpreted as the dealers' aggregate inventory.

²⁰ In Section 4.3.2, we analyze the imperfectly competitive case in which dealers take account of the impact of their orders on the clearing price (i.e., exert market power). But as we explain there, the effects of inventory risk on price dynamics, our topic in this section, do not qualitatively depend on whether dealers are competitive or not.

prices respond not only to the current order flow but also to the anticipated order flow, so that in equilibrium the pattern of orders, prices, and dealers' inventories over time is jointly determined.

3.5.1 A Static Model: Inventory Risk and Liquidity

Let us first consider a static setting, where trading takes place at time $t = 0$ and the asset pays out its fundamental value v at time $t = 1$. Thus dealers value their final inventory position z_1 at $v \cdot z_1$. Using expression (3.41), their final wealth is:

$$w_1 = v(z_0 - y_0) + c_0 + p_0 y_0. \quad (3.42)$$

Since dealers are risk averse, their objective function is increasing in the expected value of their end-of-period wealth $E_0(w_1)$ and decreasing in its riskiness, which we measure by its variance $\text{var}_0(w_1)$.

The most commonly used formulation of the objective function of risk-averse investors (and one that will be used again in Chapter 4) is the linear mean-variance function:

$$U = E_t(w_{t+1}) - \frac{\rho}{2} \text{var}_t(w_{t+1}), \quad (3.43)$$

where ρ is a measure of risk aversion. Computing the objective (3.43) for the dealer's end-of-period wealth in expression (3.42) and denoting the variance of ε_1 by σ_ε^2 , we get:

$$\begin{aligned} U &= E_0(v)(z_0 - y_0) + c_0 + p_0 y_0 - \frac{\rho}{2} (z_0 - y_0)^2 \sigma_\varepsilon^2 \\ &= \mu_0(z_0 - y_0) + c_0 + p_0 y_0 - \frac{\rho}{2} (z_0 - y_0)^2 \sigma_\varepsilon^2, \end{aligned}$$

recalling that $E_0(v) \equiv \mu_0$. The dealer chooses his supply of shares y_0 so as to maximize this objective function. The dealer's first-order condition with respect to y_0 is:

$$\frac{\partial U}{\partial y_0} = -\mu_0 + p_0 + \rho(z_0 - y_0)\sigma_\varepsilon^2 = 0,$$

which yields his inverse supply function (i.e., the price at which he is willing to supply a given amount of shares):

$$p_0 = \mu_0 + \rho\sigma_\varepsilon^2(y_0 - z_0). \quad (3.44)$$

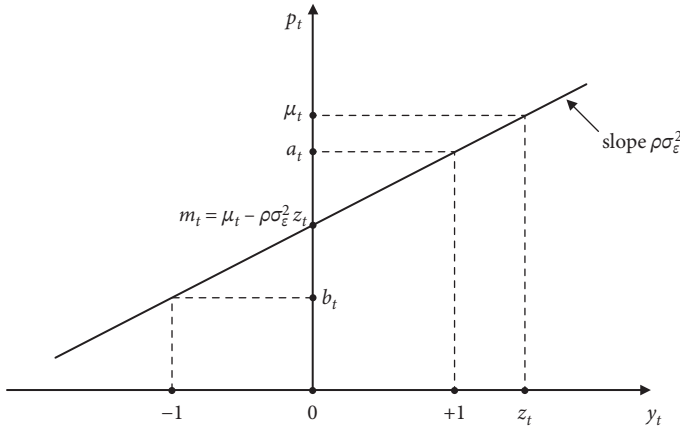


Figure 3.5 Dealers' supply and quotes in the static model with risk-averse dealers.

Thus, as shown in Figure 3.5, if the price equals the fundamental value $E_0(v) = \mu_0$, so that the market offers no compensation for inventory holding costs, the dealer supplies his entire initial inventory z_0 , so as to reduce his end-of-period inventory ($z_1 = z_0 - y_0$) to zero. But if the market offers a higher price and therefore a risk premium, the dealer is willing to supply more shares, thus taking a risky short position. The size of the short position depends on the dealer's degree of risk aversion ρ and on the stock's fundamental volatility σ_ϵ^2 : the more risk averse the dealer and the riskier the stock, the fewer shares the dealer will supply. Conversely, if the price is below the fundamental value, the dealer will supply less than his initial inventory: if shares trade at a discount relative to their fundamental value, he is willing to hold a long position until time 1.

In equilibrium, the representative dealer must supply exactly the amount of shares needed to satisfy the incoming order flow, that is, $y_0 = q_0$, where q_0 is the number of shares demanded by investors. Replacing this condition in the dealer's supply function (3.44) yields the equilibrium price:

$$p_0 = \underbrace{\mu_0 - \rho\sigma_\epsilon^2 z_0}_{\text{midquote } m_0} + \rho\sigma_\epsilon^2 q_0 = m_0 + \rho\sigma_\epsilon^2 q_0. \quad (3.45)$$

Hence the equilibrium midquote m_0 reflects not only the stock's expected fundamental value, but also an inventory risk adjustment: it is the price at which the dealer is willing to hold precisely his initial inventory. As shown in Figure 3.5, the equilibrium price, p_0 , depends on whether the dealer receives a buy or a sell order. The ask price for a buy order of size $|q_0|$ exceeds the midquote, reflecting the need to reward the dealer for supplying shares out of his inventory, while the

bid price for a sell order of the same size incorporates a discount to reward him for adding one extra unit to it:

$$p_0 \equiv \begin{cases} a_0 = m_0 + \rho \sigma_\varepsilon^2 |q_0| & \text{if } q_0 > 0, \\ b_0 = m_0 - \rho \sigma_\varepsilon^2 |q_0| & \text{if } q_0 < 0. \end{cases} \quad (3.46)$$

Figure 3.5 illustrates the ask and bid prices for an order of unit size ($q = 1$). Hence the bid-ask spread is determined by the dealers' risk aversion ρ , the stock's fundamental volatility σ_ε^2 and the size of the trade q , in other words, by inventory holding costs:

$$S = 2\rho\sigma_\varepsilon^2 q. \quad (3.47)$$

Chapter 4 will further examine the determination of the bid-ask spread in the presence of inventory risk and specifically how market liquidity is affected by the number of dealers competing in the market.

3.5.2 A Dynamic Model: Anticipated Inventory and Prices

A limitation of the static model in the previous section is that it does not take into account that dealers can manage the inventory risk arising from filling orders by unwinding the resulting change in inventory over time. To capture this point, in this section we present a multi-period setting where prices reflect both current inventories and the time path of future inventories, that is, prices react both to the current and the expected order flow. Then, both the immediate price impact of an order and its subsequent persistence depend on how long the resulting change in inventories will last before being unwound by offsetting orders.

These holding-period effects can be simply captured by extending the static model of the previous section to a setting with two trading periods, 0 and 1, where dealers receive orders in both periods and engage in dynamic inventory management. Hence, at $t = 0$ they set their quotes in anticipation of future order flow at $t = 1$. For tractability, the future order flow is assumed to be fully anticipated, that is, dealers already know q_1 when they trade at time 0. At time $t = 2$, the representative dealer liquidates his position in the risky asset at the fundamental value $v = \mu_0 + \varepsilon_1 + \varepsilon_2$, where ε_1 and ε_2 are the two innovations in the fundamental value of the security that become known immediately after trading at $t = 0$ and $t = 1$, respectively.

The dealer's final wealth at $t = 2$ is the sum of his cash and inventory position:

$$w_2 = vz_2 + c_2 = v(z_1 - y_1) + c_1 + p_1 y_1, \quad (3.48)$$

where in the second step the final inventory z_2 is re-expressed using (3.39) and the final cash balance c_2 using (3.40). Repeating the process again for z_1 and c_1 , the dealer's final wealth can be rewritten as:

$$w_2 = v(z_0 - y_0 - y_1) + c_0 + p_0 y_0 + p_1 y_1, \quad (3.49)$$

which can be rearranged to re-express the dealer's final wealth as the sum of the liquidation value of his initial position and his trading profits at $t = 0$ and $t = 1$:

$$w_2 = (v z_0 + c_0) + (p_0 - v) y_0 + (p_1 - v) y_1. \quad (3.50)$$

The dealer is facing a dynamic optimization problem, where in choosing supply at time 0, y_0 , he needs to anticipate his trading strategy at time 1, y_1 . The way to solve problems of this type is by backward induction: first, solve for the optimal trading strategy at time 1 for any given positions inherited from time 0; then, work back in time to find the optimal strategy at time 0, correctly anticipating subsequent optimal choices at time 1 and their impact on trading profits at time 1.

In the first step, i.e., the choice of optimal supply at $t = 1$, y_1 , the dealer faces the same static problem analyzed in Section 3.5.1. Thus the equilibrium price at $t = 1$ is given by applying expression (3.45) to time 1 (rather than time 0):

$$p_1 = \mu_1 - \rho \sigma_\varepsilon^2 z_1 + \rho \sigma_\varepsilon^2 q_1. \quad (3.51)$$

Hence, the dealers' trading profits at $t = 1$ are:

$$(p_1 - v) y_1 = (\mu_1 - \rho \sigma_\varepsilon^2 z_1 + \rho \sigma_\varepsilon^2 q_1 - v) y_1. \quad (3.52)$$

The market clears in both periods: $y_1 = q_1$ and $z_1 = z_0 - q_0$. Hence, recalling that $v = \mu_1 + \varepsilon_2$, the equilibrium value of trading profits at $t = 1$ is:

$$(p_1 - v) q_1 = [\rho \sigma_\varepsilon^2 (\underbrace{z_0 - q_0 - q_1}_{z_2}) + \varepsilon_2] (-q_1). \quad (3.53)$$

This expression shows that the dealer's profits on his trading at $t = 1$ equal the number of shares purchased at $t = 1$ (that is, $-q_1$) multiplied by their return, which in turn consists of a risk premium compensating the dealer for holding the final inventory z_2 plus the final shock to the fundamental value ε_2 .

Inserting this expression for the profits earned at $t = 1$ into equation (3.50) yields the dealer's final wealth, given equilibrium at $t = 1$:

$$w_2 = \nu z_0 + c_0 - \rho \sigma_\varepsilon^2 (z_0 - q_0 - q_1) q_1 + (p_0 - \nu) y_0 - \varepsilon_2 q_1. \quad (3.54)$$

Hence, the dealer's objective function at $t = 0$ is:

$$U = E_0(w_2) - \frac{\rho}{2} \text{var}_0(w_2), \quad (3.55)$$

where the expected wealth is:

$$E_0(w_2) = \mu_0 z_0 + c_0 - \rho \sigma_\varepsilon^2 (z_0 - q_0 - q_1) q_1 + (p_0 - \mu_0) y_0, \quad (3.56)$$

and its variance is:

$$\text{var}_0(w_2) = [2(z_0 - y_0)^2 + q_1^2 - 2(z_0 - y_0) q_1] \sigma_\varepsilon^2, \quad (3.57)$$

recalling that $\text{var}_0(\nu) = \text{var}_0(\varepsilon_1 + \varepsilon_2) = 2\sigma_\varepsilon^2$.

The first-order condition of the dealer's objective function with respect to y_0 yields his optimal supply of the security at $t = 0$:

$$y_0(p_0) = \frac{p_0 - \mu_0}{2\rho\sigma_\varepsilon^2} + z_0 - \frac{q_1}{2}. \quad (3.58)$$

The first two terms of this expression are the supply that the dealer would choose at $t = 0$ if he did not expect to trade again at $t = 1$. The third term is new, and captures the dealer's pre-positioning in anticipation of future order flow: if the dealer expects clients to demand an amount $q_1 > 0$ in the future, he anticipates that his inventory is going to be depleted in the future, and therefore replenishes it at $t = 0$ by half of that amount.

In equilibrium, $y_0 = q_0$, and therefore the price is:

$$p_0 = \mu_0 + \rho \sigma_\varepsilon^2 [2(q_0 - z_0) + q_1]. \quad (3.59)$$

Hence, the initial price reacts not only to current demand for the security but also to future demand, because the inventory management by dealers in anticipation of future demand transfers some of its pressure on the initial price. Otherwise stated, the market price reflects the fact that a position of $z_0 - q_0$ will have to be held for two periods and a position of $-q_1$ will have to be held for one period.

In reality, orders are serially correlated, and thus at $t = 0$ market makers anticipate that the current order q_0 will trigger a subsequent order q_1 . To capture this idea in a simple way, assume that $q_1 = \delta q_0$, where positive autocorrelation ($\delta > 0$) may result from the gradual trickling of large orders, while negative

autocorrelation ($\delta < 0$) can result from market participants' reaction to attractive prices quoted by market makers ($p_0 \neq \mu_0$).

The resulting bid and ask prices are:

$$p_0 \equiv \begin{cases} a_0 = m_0 + \rho\sigma_\varepsilon^2(2 + \delta)|q_0| & \text{if } q_0 > 0, \\ b_0 = m_0 - \rho\sigma_\varepsilon^2(2 + \delta)|q_0| & \text{if } q_0 < 0, \end{cases} \quad (3.60)$$

where $m_0 = \mu_0 - 2\rho\sigma_\varepsilon^2 z_0$. So, in the baseline case where the order flow is not serially correlated ($\delta = 0$), the bid-ask spread is doubled compared to the static model (see equation (3.46)), because the inventory risk generated by the absorption of the order flow lasts for two periods. The prospect that the dealer will receive offsetting orders in the future ($\delta < 0$) mitigates his inventory holding costs, and therefore the bid-ask spread. In contrast, if the order flow is positively autocorrelated ($\delta > 0$), the actual holding costs stemming from a given order is larger than in the baseline case. Hence, the size of the bid-ask spread depends on the autocorrelation of the order flow, δ .

In terms of price dynamics, a key implication of this model is that the future order flow, not just the current one, contributes to determine price movements via its effect on dealers' inventories. To see this, use equations (3.59) and (3.51) to write the expected change in the equilibrium price between $t = 0$ and $t = 1$:

$$E_0(p_1) - p_0 = \rho\sigma_\varepsilon^2 \underbrace{(z_0 - q_0)}_{z_1}, \quad (3.61)$$

that is, the expected return on the dealer's position between time 0 and time 1 is per-period risk premium on his inventory position held over this period. Similarly, the expected price change between $t = 1$ and $t = 2$, from equation (3.51), is:

$$E_0(p_2 - p_1) = E_0(v - p_1) = \rho\sigma_\varepsilon^2(z_1 - q_1) = \rho\sigma_\varepsilon^2(z_0 - (1 + \delta)q_0). \quad (3.62)$$

Figure 3.6 illustrates the expected price dynamics in the wake of a sell order, i.e., for $q_0 < 0$. For simplicity, in the figure it is assumed that the inventory of dealers is zero before the arrival of the order, both at $t = 0$ ($z_0 = 0$) and before, as no order arrives in the previous period $t = -1$. Hence before the arrival of the order at $t = 0$, the price is simply the asset's risk-neutral valuation: $p_{-1} = \mu$. The price dynamics after the sell order depends on the autocorrelation δ in the order flow. In line with equation (3.60), the price drop and its subsequent reversal are largest if the order flow is positively correlated over time ($\delta > 0$), intermediate if it is uncorrelated ($\delta = 0$) and mildest if it is negatively autocorrelated ($\delta < 0$). Intuitively, if orders are positively correlated over time, then the impact of

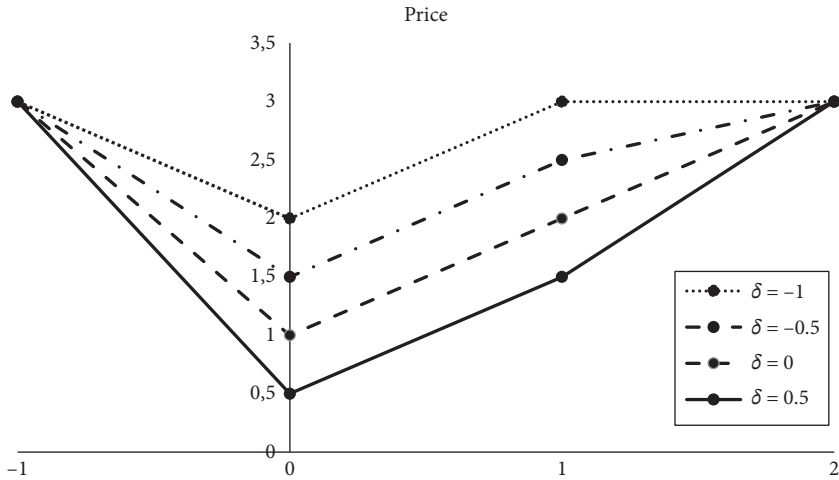


Figure 3.6 Expected prices in the dynamic model with risk-averse dealers.

current orders will be exacerbated by inventory accumulation. Conversely, if orders tend to reverse over time, the price impact of any initial order will be mitigated, because dealers are exposed to inventory risk for a shorter time. Accordingly the stronger is the reversal in the order flow, the milder is the initial price impact, as well as the subsequent reversal, as can be seen by comparing the cases where $\delta = -0.5$ and $\delta = -1$. Figure 3.7 plots the dynamics of inventories in the immediate aftermath of trades (that is, the inventory shown in the figure at any time t is z_{t+1}). Comparing Figure 3.6 and Figure 3.7 one sees that price changes anticipate subsequent fluctuations in inventories: when inventories are expected to rise more in the future, the current price drops more, and therefore both price and inventory display larger subsequent reversals.

As done in Section 3.4, one can measure the expected short-term and long-term impact of a buy order, benchmarked against the midprice at $t = 0$. For simplicity, consider a situation in which market makers start with zero inventory ($z_0 = 0$), and suppose that a buy order $q_0 > 0$ reaches the market at $t = 1$. From equation (3.60) the short-term impact of the order is:

$$\text{ST impact} = \rho \sigma_{\varepsilon}^2 (2 + \delta) q_0, \quad (3.63)$$

which is increasing in δ , i.e., in the persistence of the order flow.

Interestingly, in the medium term the order impacts the price by less than in the short term: adding the expected price change between $t = 0$ and $t = 1$ from equation (3.61) to the short-term impact from equation (3.63), the medium-term impact of the buy order on the expected price at $t = 1$ is:

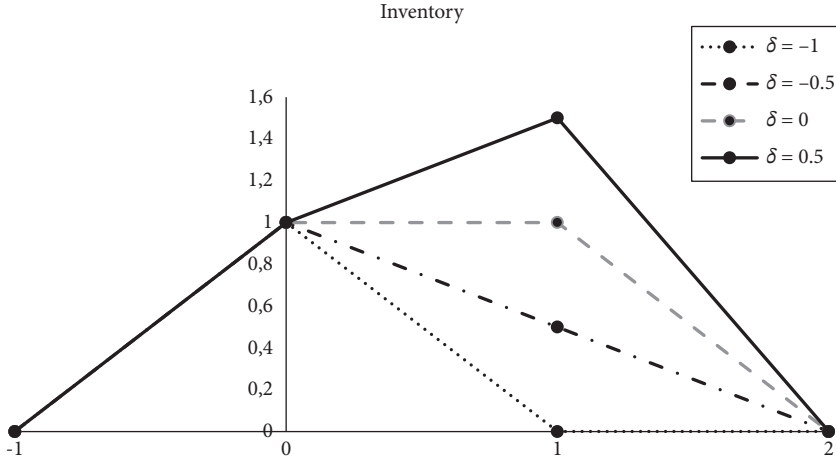


Figure 3.7 Dealers' inventories in the dynamic model with risk-averse dealers.

$$\text{MT impact} = E_0(p_1) - m_0 = \rho\sigma_\varepsilon^2(1 + \delta)q_0, \quad (3.64)$$

which is smaller in absolute size than the short-term impact, highlighting that there is mean-reversion in the expected price.

Of course, the long-term impact of the order is zero, because it is natural to assume that eventually the market makers' inventory goes back to the initial level, so that the price will go back to the initial midquote: in our model, this is implicitly assumed to happen at $t = 2$. As shown in Figure 3.6, the speed of mean reversion depends on the autocorrelation in the order flow: the expected price reverts faster to the initial level the lower is the value of δ . The reason is that, as shown in Figure 3.7, when there is offsetting order flow the market makers can unwind their inventories faster, and therefore their inventory holding cost is smaller.

An important general principle that emerges from this analysis is that the risk premium component of the market price at any moment in time is proportional to the holding time for which a given position will be held, and not only on the expected size of this position over this holding period. Specifically, if market makers anticipate a future time path of inventories $(z_{t+1}, \dots, z_T)$, then the price at any time t is equal to:

$$p_t = \mu_t - \rho\sigma_\varepsilon^2 \sum_{\tau=t+1}^{t=T} z_\tau, \quad (3.65)$$

T being the last time at which the dealer expects his inventory to differ from zero. This can easily be checked for $t = 0$ and $t = 1$ for the case with two trading dates considered in this section.²¹

3.5.3 Joint Dynamics of Prices and Inventories: Evidence

The inventory holding cost model analyzed so far yields three main empirical predictions. First, the pressure that orders exert on prices is inversely proportional to the inventory imbalance that they induce and to the persistence of such imbalance. Second, offsetting orders will gradually allow market makers to unwind these inventory imbalances, so that we should observe mean reversion in their inventories. Third, as a consequence, prices should also revert towards the initial level as inventories are unwound: indeed, the path of prices should be the mirror image of that of market makers' inventories.

Consider the first of these three predictions. When a dealer has a long position before trading at date t ($z_t > 0$), buying additional shares is not attractive, as it increases his risk exposure. This will put downward pressure on the price. Conversely, a dealer with a short position is willing to trade at a relatively high price. Hence, the mid-price at date t will deviate from the fair value of the security, and the size of this deviation ($m_t - \mu_t$) is inversely related to dealers' aggregate inventory. This deviation can be called the "price pressure" arising from inventory holding costs.

For stocks listed on the NYSE, Hendershott and Menkveld (2014) find evidence consistent with this prediction. They estimate the relationship between the midquote and dealers' inventories, and obtain a negative coefficient for the price pressure per unit of inventory. Moreover, they find that this price pressure coefficient is much greater (in absolute value) for small-capitalization than for large-capitalization stocks. Specifically, a \$100,000 inventory results in price pressure of 1.01% for small stocks and 0.02% for large stocks. This makes sense, in that fewer people are interested in trading smaller stocks, so it takes longer to unwind the inventories and dealers remain exposed to risk for a longer time. In other words, for small stocks the inventory holding period is likely to be longer than for large stocks. We have shown that price pressure is the product of three terms: the expected inventory holding period, the risk aversion coefficient and the variance of the fundamental. This can be seen by comparing the expression for the midquote in the static model ($m_0 = \mu_0 - \rho\sigma_\varepsilon^2 z_0$) with that in the dynamic model ($m_0 = \mu_0 - 2\rho\sigma_\varepsilon^2 z_0$): in the latter, the price pressure coefficient is twice as

²¹ To check it, use equations (3.59) for p_0 and (3.51) for p_1 , recalling that $q_0 = z_0 - z_1$ and $q_1 = z_1 - z_2$.

large as in the former, because inventories are held for two periods rather than than one.

The second main empirical prediction of the inventory holding cost model is that inventories are mean-reverting. Comerton-Forde et al. (2010) show that the aggregate inventories of NYSE specialists over 1994–2004 fluctuate around an average value of about \$200 million. Such fluctuations are also found for individual stocks and corporate bonds, as illustrated by Figure 3.8. The top panel of the figure plots the daily inventory of Apple stock held by a broker-dealer over one year: it is clear that inventory reverts to zero in a matter of days. The bottom panel of the figure instead refers to the corporate bond market, and plots the position of a broker-dealer in a single corporate bond. This illustrates that the speed of inventory mean reversion depends on the activity level of the corresponding market: as the corporate bond market is typically less active, it features slower mean reversion in inventories.

The speed of response of the order flow and mean reversion of inventories not only depends on the market's general level of activity, as illustrated by the comparison between the time pattern of inventories in the Apple stock and in the corporate bond market shown in Figure 3.8. It also depends on other market characteristics, such as the extent of investors' monitoring of the security, and on the salience of price discrepancies, which in turn reflect dealers' risk aversion and the stock's riskiness. Moreover, in practice, dealers actively seek to control their inventory not just by pricing but also by advertising opportunities and soliciting customer orders: they can reduce the holding time of their inventory, and thus accelerate its mean reversion, by eliciting inventory-reducing orders. In line with this prediction, researchers—such as Reiss and Werner (1998) and Hansch, Naik, and Viswanathan (1998)—have found empirically that dealers with long positions are more likely to execute buy market orders; those with short positions, sell market orders. Finally, the speed of adjustment will reflect regulatory limits on the size of dealers' positions, such as short-sales constraints or margin constraints on leverage. Intuitively, the more stringent these constraints, the more apparent the footprints of inventory holding costs on price dynamics.

The third prediction of the model is that, as inventories revert, so do prices, as illustrated in Figures 3.6 and 3.7. This is illustrated by Figure 3.9, which shows the price response to large stock orders placed by clients of a major broker-dealer and executed over three days. The graph is best interpreted as representing the effect of a large sell order, and as such is the empirical counterpart of the price pattern predicted by our dynamic model and illustrated by Figure 3.6.

The mean reversion in prices is faster in more active markets, where inventories revert faster to their desired level (as shown by Figure 3.8). This is documented by Hendershott and Menkveld (2014), who estimate the duration

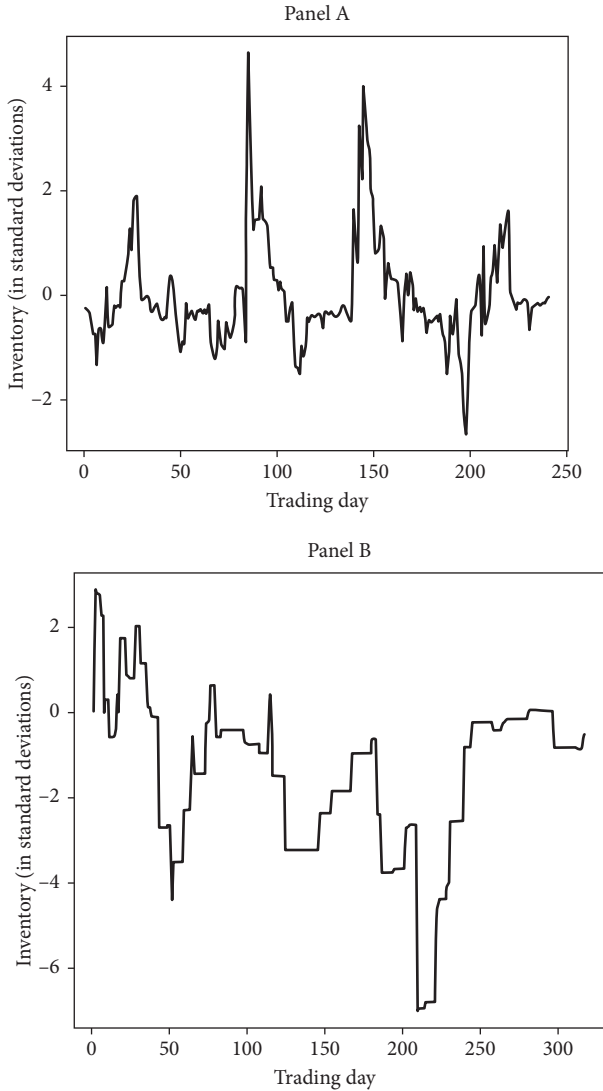


Figure 3.8 Panel A: daily inventory of the U.S.-dollar position of a block market making desk of a major broker-dealer for Apple Inc. stock, after scaling by the sample standard deviation of inventory for the sample period (2010–11). Panel B: daily inventory of the U.S.-dollar position of a market making desk of a major broker-dealer for a single investment-grade corporate bond, after scaling by the sample standard deviation of inventory levels for the sample period (2010–11). Sources: fig. 1 and fig. 3 from Duffie (2012).

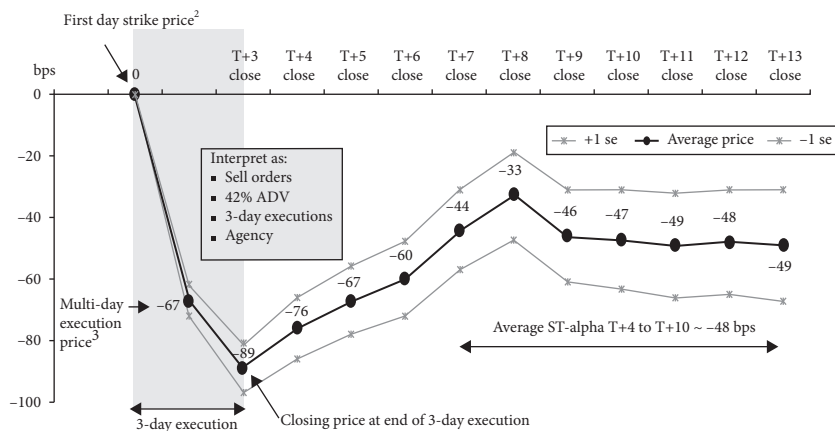


Figure 3.9 Value-weighted average closing prices from when a multi-day execution completes through ten days later. Prices are measured relative to the first-day order arrival price. The sample includes both buys and sells, but the sign of buys is flipped. *Source:* Exhibit 6 from Cai and Sofianos (2006).

of price pressure for NYSE stocks, and find that price pressure lasts longer for small-capitalization stocks (besides being stronger), presumably because trading is less frequent.²²

Mean reversion in security prices is often observed not only as a result of large orders, as in Figure 3.9, but also when security markets are hit by major shocks, such as the deletion from a widely traded stock market index or a regulatory intervention that forces some intermediaries into fire sales of certain securities, such as low-grade bonds (see Chapter 11).

Box 3.2 Dealers' Inventories, Liquidity, and Volatility Before and After The 2008–9 Crisis

The inventory holding cost model shows that dealers' willingness to accumulate and decumulate inventories in order to serve investors' orders is a key to the market's ability to provide liquidity, and that this willingness depends on the volatility of the fundamentals. So it is no surprise that, with the increase in risk during the financial crisis of 2008–9, dealers' willingness to hold inventories diminished. Their ability to hold large inventories was further impaired by regulatory restrictions on their proprietary trading. Nicole Bullock's article

²² They also measure the contribution of price pressure induced by inventory holding costs to daily volatility, finding that it ranges from 0.17% for large stocks to 1.20% for small stocks.

“Wall Street: inventory reductions ‘the death of trading’” (*Financial Times*, March 12, 2012) argues that this has reduced liquidity and increased the volatility of prices in the corporate bond market, as the model analyzed above predicts:

Before the global financial crisis, large Wall Street dealers used to hold big inventories of corporate bonds to facilitate trading for their clients.

When a bond manager such as Jesse Fogarty, of Cutwater Asset Management, wanted to sell a large number of bonds, the dealer would sell some of them to other clients for him and buy some itself to sell down over time, hopefully for a profit.

Since the financial crisis, however, large Wall Street dealers have sharply cut the number of corporate bonds they hold to facilitate this kind of trading for clients or for proprietary positions.

Mr. Fogarty calls this move “the death of trading.”

Known in the financial markets as “dealer inventories,” these bond holdings reached highs of more than \$200bn in 2007, but dropped to less than half that by the end of 2008, as dealers reduced their risk because of the banking crisis.

In the second half of last year, dealers cut their bond holdings again, as they dropped from about \$90bn at the start of June to \$40bn by the end of February 2012, the lowest level in a decade.

These lower inventories, analysts and investors say, reflect an environment of lower risk-taking by Wall Street banks in the aftermath of the financial crisis and ahead of tighter regulations.

The result is likely to be higher volatility in bond prices, as the buffer is far smaller now.

In addition, the so-called Volcker Rule, a part of the 2010 Dodd Frank Act, could make it impossible for dealers to take these kinds of positions, because it prohibits proprietary trading by banks. At the same time, higher capital costs under Basel III also loom as an impediment to holding bonds on a bank’s balance sheet. Many in the market believe the rules, which still need to be finalised, are a big driver of the reduction in inventory that was seen last year.

[...] market participants generally expect lower inventories will increase volatility, as they say happened during last year’s sell-off.

As Europe’s debt crisis escalated, risk premiums on investment grade and high-yield bonds widened sharply.

“The takeaway is that market liquidity has been reduced,” says Chris Taggart, analyst at CreditSights, a research group. “You end up with higher

volatility for the simple fact that, without the dealers taking positions, trading volumes will move prices faster.”

3.6 The Full Picture

In this chapter, we have seen that an order's short run and long run effects on transaction prices differ depending on what determines the bid-ask spread, that is, whether the source of market illiquidity is adverse selection costs, order-processing costs or rents, or inventory holding costs. In this section we bring these factors together, illustrating qualitatively the type of price response to the order flow that we should observe depending on which of the three underlying forces determines it. Figure 3.10 plots the expected path of the transaction price over time, conditional on a buy order coming to the market at time t . The reason why we plot the expected rather than the actual price path is that, at any time, transaction prices are affected by news on fundamentals as well as by new orders, so that the actual price path is subject to continuous shocks that may conceal the effect of the buy order at time t . By taking expectations of the price at time

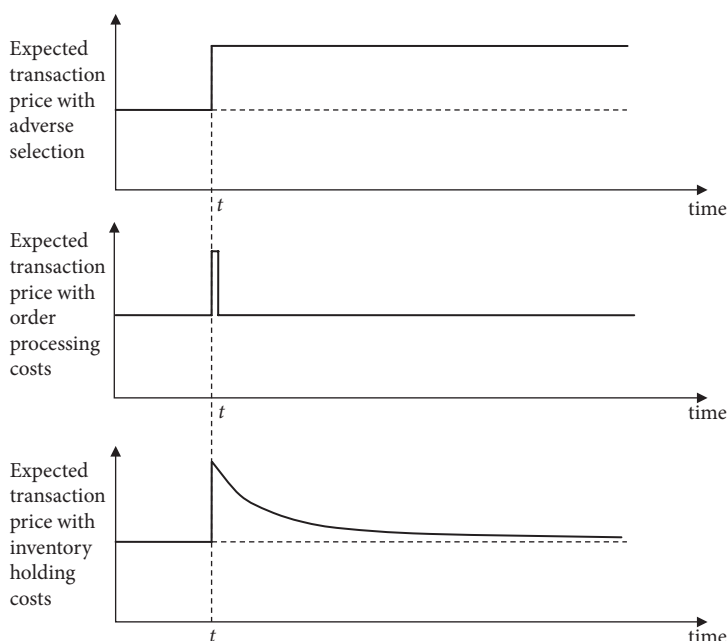


Figure 3.10 Price reaction to a market buy order in different settings.

t , we filter out this noise and highlight the “impulse response” to the buy order submitted at that time.

The top panel of the figure shows that in a setting with adverse selection the price impact of an order is permanent: the buy order induces traders to revise their beliefs about fundamentals upwards and embed this change in beliefs fully and permanently in their quotes. By the same token, in this setting there is no difference between the short run and the long run effect of the order. The middle panel shows that with order-processing costs (but no adverse selection) the buy order induces a temporary upward blip in the transaction price: the effect is present only in the very short run and disappears in the longer term, generating short-term negative autocorrelation in transaction price changes. Merging the time impacts illustrated in the top and middle panels, we have the short-term and long-term price reactions described by equations (3.34) and (3.36), which merge the adverse selection and order-processing cost models.

The bottom panel displays the price path when a buy order is submitted to risk averse dealers: the price reaction is positive in the short term but trails off in the medium term, as shown by equations (3.63) and (3.64). So there is a price reversal as in the middle panel, but the price impact dissipates slowly over time, as the inventories are gradually brought back to target. So again in this case returns should exhibit negative autocorrelation, but at lower frequencies and for longer lags than in the case of order-processing costs.

In the real world, of course, all three effects may be present at once. This is illustrated in Figure 3.11, which shows how all three may contribute to some extent to the positive short-term impact of a buy order, with some reversal occurring quickly as a result of order-processing costs, some further reversal occurring more gradually due to inventory holding costs, and in the long run only the informational effect associated with adverse selection persisting.

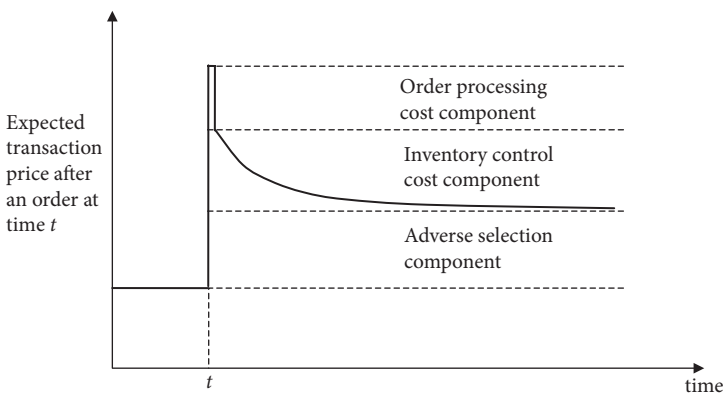


Figure 3.11 Short-term and long-term price responses to a market buy order.

Market microstructure researchers have repeatedly sought to decompose the bid-ask spread into the three components of adverse selection, order-processing costs (plus rents), and inventory holding costs. To identify these components of market illiquidity, researchers exploit the fact that each one has a different effect on the time-series properties of prices and orders, as is shown in the previous figure. Chapter 5 will explain in detail the econometric techniques that are applied to identify these effects and measure the contribution of each source of market illiquidity to the bid-ask spread.

3.7 Further Reading

The model of trading with asymmetric information developed in Section 3.3 is based on Glosten and Milgrom (1985). This model has been extended in many directions. For instance, Easley and O'Hara (1987) consider the possibility of multiple trade sizes for the informed investor; Easley and O'Hara (1992) introduce the possibility of information event uncertainty (that is, the possibility of there being no change in the asset value). In this case, dealers learn not only from the order flow but also from trading volume (the total number of trades over a given period).

One important feature of this model is that liquidity suppliers have less information than some liquidity demanders. In a model related to Glosten and Milgrom (1985), Calcagno and Lovo (2006) provide a theoretical analysis of price competition among dealers when some dealers are better informed than others. In this case, dealers' quotes are informative; the order flow not. Bloomfield, O'Hara, and Saar (2005) show experimentally that informed investors optimally use both market and limit orders when they have the requisite flexibility to do so. Overall, Calcagno and Lovo (2006), the experimental evidence in Bloomfield, O'Hara, and Saar (2005) and empirical evidence in Brogaard, Hendershott, and Riordan (2019) suggest that in securities markets quotes also contain information, in addition to order flow.

The model by Glosten and Milgrom (1985) has also been used to address various policy issues, as we do in Chapter 8. For instance, Diamond and Verrecchia (1987) use it to study the effects of short-sales constraints on price discovery. They show that transaction prices remain efficient in the semi-strong form with short-sales constraints, as in Glosten and Milgrom (1985), but that short-sales constraints slow the process of price discovery. Intuitively, such constraints prevent informed investors from selling a stock when they have bad news, which eventually reduces the speed at which prices reflect informed investors' private information.

Stoll (1978) was the first to relate the bid-ask spread to inventory holding costs. Ho and Stoll (1981) analyze the optimal dynamic pricing policy for a monopolist dealer in presence of inventory risk (see also Madhavan and Smidt, 1993; Hendershott and Menkveld 2014). Amihud and Mendelson (1980) also consider a dynamic model in which dealers are risk neutral but face position limits (i.e., a constraint on the maximum size of their inventory). They show that position limits also generate mean reversion in inventories. Ho and Stoll (1983), Biais (1993), and Yin (2005) study competition between dealers when they have different inventories in various market structures. The empirical literature exploits the idea that inventory effects dissipate in the long run to separate adverse selection from inventory holding costs (see Chapter 5). Given inventory risk, information on the aggregate inventories is informative on the short-term path of prices. Such information may also be a source of adverse selection; its effect is analyzed by Vayanos (1999).

The speed of dealers' inventories' reversion to the mean is sometimes used by researchers as a gauge of the importance of inventory risk in securities markets (see, for instance, Hasbrouck and Sofianos 1993; Madhavan and Smidt 1993; Lyons 1995; Comerton et al. 2010; and Hendershott and Menkveld 2014). Interestingly, the rate of reversion to the mean varies across assets. For instance, Madhavan and Smidt (1993) and Hasbrouck and Sofianos (1993) empirically find a slow rate of mean reversion for the inventories of the specialists in NYSE stocks, whereas Lyons (1995) finds a very high rate of reversion for a dealer in the Deutsche Mark/Dollar market. These differences may reflect differences between assets in price volatility, risk aversion of dealers, or responsiveness of order flow. Recent years have seen the emergence of high-frequency market-makers. These market-makers automate the posting of their quotes and use their speed of access to various trading platforms and information to manage inventory risk very efficiently. As a result, they hold their inventories for a very short period of time (see, for instance, evidence in Menkveld 2013). This suggests that the speed of reversion to mean in dealers' inventories and prices may have increased considerably in recent years.

3.8 Exercises

1. **Bid-ask spread with order-processing costs.** In Section 3.4.1 order-processing costs are assumed to be γ per share traded. Consider the following alternative assumptions:
 - a. Assume that order-processing costs are k per transaction. Compute the bid-ask spread in this case and show that it is decreasing with the size

- of the transaction. Which features of the technology of trading would lead you to think that this is a realistic model of order-processing costs?
- b. Assume that order-processing costs are k per euro traded. Show that the absolute bid-ask spread is increasing in the security's underlying value and the relative bid-ask spread is constant, in contrast with the expressions found in the text where order-processing costs are a constant γ per euro traded, irrespective of the share value.
 - c. Stoll (2000) reports that Roll's measure ranges from 6.45 cents for small-company NASDAQ stocks to 13.17 cents for large-company NASDAQ stocks (p. 1494). Note that, on average, small-company stocks have a lower price. Which of the following hypotheses is more consistent with this empirical evidence?
 - (i) Order-processing costs are constant per share traded.
 - (ii) Order-processing costs are constant per dollar of value traded.
- 2. Bid-ask spread and insider trading.** A small risky company's stock is worth either \$10 (v^L) or \$20 (v^H) with probability $\frac{1}{2}$ each ($\theta = 1 - \theta = 0.5$).
- a. Compute the bid and ask prices set by risk neutral competitive market makers in the absence of informed trading.
 - b. Compute the bid and ask prices set by risk neutral competitive market makers when they expect one in ten of trade initiators to be informed (to know the stock's true value) and to trade as profit maximizers, while the other nine out of ten are uninformed and buy or sell with equal probability. Assume that all transactions are of the same size.
 - c. Compute the average trading cost to an uninformed trader and the average gain to an informed one, assuming a unit trade size in both cases.
 - d. Do you agree with the following statement? "Insider trading does not harm most market participants: it harms only those who are unlucky enough to trade with an insider." Why? Refer to the example in this exercise to illustrate your argument.
- 3. Insiders' expected profits in the Glosten-Milgrom model.** Consider the static version of the Glosten-Milgrom model, where market makers set the first ask and bid equilibrium quotes of the trading day. Compute the expression for the expected profits of the insider (only as function of the parameters π , v^H , and v^L). In computing this expression, take into account that the informed trader can trade both in the state in which the security is worth v^H and in that in which it is v^L , and that each of these two states has probability $\frac{1}{2}$. How does this expression vary as a function of the probability of informed trading, π ? What is the intuitive rationale for this result?

4. Imperfectly informed investors in the Glosten-Milgrom model. Consider the one-period Glosten-Milgrom model, where the security's true value v can be high (v^H) or low (v^L) with probability $\frac{1}{2}$ each. Market makers are competitive and risk neutral, and do not know v . In each period, a single trader comes to the market: with probability $1 - \pi$, he is a noise trader, who buys or sells one unit with probability $\frac{1}{2}$ each; with probability π he is an informed trader, who observes a signal about security's true value. With probability $\rho \in \left(\frac{1}{2}, 1\right]$ the signal is accurate, that is, it coincides with the true value of the security. With probability $1 - \rho$, instead, the signal is mistaken, so that the insider assigns the wrong value to the security. Hence, ρ measures the accuracy of the signal observed informed trader: for ρ close to $\frac{1}{2}$, the informed would be similar to a noise trader; for $\rho = 1$, the insider trader would be perfectly informed.

- a. Write down dealers' expected profits when they receive (i) a buy and (ii) a sell order. [Hint: assume that the informed trader buys the security when his signal equals v^H and sells when it is v^L , so that in each instance with probability ρ he makes profits and with probability $1 - \rho$ he makes losses.]
- b. Compute the bid and ask prices set by risk neutral competitive market makers.
- c. Derive the bid-ask spread as a function of signal's informativeness. How does this result compare with the case of perfectly informed insider trading? Is the market more or less illiquid? Intuitively, why?
- d. Verify whether, given the bid and ask prices derived at point **b**, the insider is actually willing to buy when his signal equals v^H and sells when it is v^L , that is, whether this strategy yields positive expected profits in equilibrium.

5. Endogenous information acquisition by insiders in the Glosten-Milgrom model. Consider the one-period Glosten-Milgrom model, where the security's true value v can be high (v^H) or low (v^L) with probability $\frac{1}{2}$ each. Market makers are competitive and risk neutral, and do not know v . In each period, a single trader comes to the market: with probability $1 - \pi$, he is a "noise trader," who buys or sells one unit with probability each; with probability π he is a "potential insider," who learns the security's true value v if he pays a cost c , in which case he will trade on his information to make a profit. (If he does not elect to acquire information, he does not trade.)

- a. Compute the bid and ask prices and the bid-ask spread that market makers set, assuming that they believe the insider will acquire information with some given probability $\varphi \in [0, 1]$?

- b. Given these prices, determine the trading profit of an insider who has decided to acquire information.
 - c. Determine the condition on the cost parameter c in terms of π and $v^H - v^L$ under which the potential insider will never choose to acquire information, even in the most auspicious situation in which market makers do not expect to face any insider trading (i.e., $\varphi = 0$).
 - d. Determine the condition on the cost parameter c in terms of π and $v^H - v^L$ under which the potential insider will always choose to acquire information, even in the least auspicious situation in which market makers expect to face insider trading with probability π (i.e., $\varphi = 1$).
 - e. Now consider intermediate values of c for which neither of the conditions determined under points **c** and **d** is satisfied. Determine the value of φ that will make the potential insider indifferent between acquiring and not acquiring information. This endogenous value of φ describes a “mixed-strategy equilibrium” in which the insider randomizes between the two options. How does this value of φ depend on the parameters of the model? Explain your results intuitively.
 - f. Characterize the equilibrium in the three ranges of the values of c considered in points **c**, **d**, and **e**. Plot the bid-ask spread and the potential insider’s net expected profit as a function of the cost c of acquiring information. Explain your graph intuitively.
- 6. Equilibrium with price-sensitive uninformed trading.** Consider the one-period version of the model developed in Section 3.3, assuming $v^H = 1$ and $v^L = 0$ with equal probabilities, and the fractions of both informed and uninformed investors are π and $1 - \pi$, respectively. Uninformed investors refuse to trade at a price that is more than δ away from the public estimate μ of the security’s value.
- a. Compute the zero-expected-profit bid and ask quotes posted by the dealers.
 - b. Plot how bid and ask prices and the probability of trading vary with π .
- 7. Market breakdown in the face of excessive informed trading.** Risk-neutral competitive market makers set bid and ask prices for a security whose final liquidation value v is distributed on $(0, \infty)$ with density:

$$f(v) = \frac{2}{(v + 1)^3}.$$

The liquidation value of the security becomes publicly known after trading; but during trading, it is known only to informed traders who come into the market with probability π and trade a unit to maximize their profit. Noise traders come with probability $1 - \pi$; they buy a unit

of the security if their personal valuation is higher than the ask price set by market makers and sell a unit if their personal valuation is lower than the bid price. Noise traders' personal valuations for the security are also distributed with the same density $f(\cdot)$, though independently of the true value v .

Compute equilibrium bid and ask prices, and show that there is a critical value of π beyond which the market breaks down, in the sense that the market makers are unable to set a profitable ask price.

In solving this exercise, you can use the fact that the cumulative distribution function corresponding to density function $f(\cdot)$ is:

$$F(v) = 1 - \frac{1}{(v+1)^2},$$

and also that for this distribution, for any given ask price a and bid price b ,

$$\begin{aligned} E(v) &= 1, \\ E(v \mid v \geq a) &= 2a + 1, \\ E(v \mid v \leq b) &= \frac{b}{b+2}. \end{aligned}$$

8. **Differences in price pressure across stocks.** Can you explain the empirical finding by Hendershott and Menkveld (2014) that the price pressure per unit of inventory is much greater for small-capitalization than for large-capitalization stocks, based on the inventory holding cost model presented in this chapter? How?
9. **Short-sale constraints, liquidity, and market efficiency (inspired by Diamond and Verrechia 1987).** Consider the market for a risky security with uncertain final value v , which can be 1 or 0 with equal probability. As in this chapter, quotes for unit trade size are posted by risk neutral market-makers, and market orders are submitted either by informed traders, who arrive with probability π and know the true value of the stock, or by liquidity traders, who arrive with probability $1 - \pi$ and wish to buy or sell with equal probability.

Traders may be subject to constraints on short selling: an informed trader is short-sale constrained with probability κ_i , and a liquidity trader is short-sale constrained with probability κ_l . If a trader is constrained, he cannot sell when he would like to do so and in this case does not trade. Let d be the direction of the market order: $d = 0$ (*no trade* if the trader wants to sell and is short-sale constrained), $d = +1$ (buy market order) and $d = -1$ (sell market order).

- a. Draw the tree for the order arrival process in this model.
- b. Calculate the bid and ask quotes, and show that the bid-ask spread does not depend on κ if short-sale constraints are identical for liquidity traders and informed traders, that is, if $\kappa_i = \kappa_l = \kappa$. What happens to the bid-ask spread if $\kappa_i > \kappa_l$ or if $\kappa_i < \kappa_l$? What is the economic intuition for these results?
- c. Compute dealers' expectations of the value of the security *if there is no trade*. Show that it is strictly lower than the unconditional expected value of the security if and only if $\kappa_i > 0$. What is the intuition for this finding?
- d. Assume that $\kappa_i = \kappa_l = \kappa > 0$. Let $p(d) = E(v | d)$ be the "price" of the security at time 1 for $d \in \{1, 0, -1\}$. (Note that $p(0)$ is not observable in the form of a quote, because it is the market maker's valuation of the security on receiving no order.) Show that:

$$E(p(d)) = E(v)$$

and that:

$$E(p(d) | d \neq 0) > E(v) \text{ if } \kappa > 0.$$

- e. Suppose that you have data for the transaction prices for stocks traded in a market before the removal of short-sale constraints (time $t-1$) and after its removal (time t). According to the model presented above, would you expect, on average, to find a negative, zero, or positive return around the date in which the constraint is removed? Can we use this result to evaluate the merits of short-sale constraints?
- 10. Predicting future order flow.** In Section 3.4.2, we discuss how the direction of the order flow at future dates t and beyond can be predicted given the current belief θ_{t-1} about the probability of a high value of the stock:
- a. Use Table 3.2 to determine $E(d_{t+T} | \Omega_{t-1})$ for any $T \geq 0$, given the current belief θ_{t-1} .
 - b. Use equation (3.16) to update beliefs in the wake of a buy order and thus compute:

$$E(d_{t+T} | \Omega_{t-1}, d_t = 1) - E(d_{t+T} | \Omega_{t-1}).$$

- 11. Empirical analysis of the determinants of spreads and the Amihud illiquidity measure.** The data for this exercise are contained in the Excel file Ch3_ex11_data.xls or in the Stata data file Ch3_ex11_data.dta available on the companion website for this book. These files provide

a record of the average values of the following variables, for a sample of 1,128 stocks traded in U.S. stock markets (based on daily data for April 2009):

- ticker code that identifies each stock (*ticker*);
 - bid price (*pb*), ask price (*pa*), and closing price (*p*);
 - number of shares (in thousands) traded per day (*vo*);
 - trading volume (in millions of dollars) per day (*vp*);
 - number of shares (in thousands) outstanding (*ibnosh*);
 - sector according to GICS classification (*gics*);
 - relative bid-ask spread, in percent (*bas100*);
 - Amihud illiquidity measure (*ami100*);
 - market capitalization, in millions of dollars (*mktcap*);
 - daily return (*ret*);
 - realized volatility (twenty-trading-day moving standard deviation of returns, *vola*).
- a. Compute the correlation matrix of the closing price, trading volume, market capitalization, bid-ask spread, average return, and return volatility. Which variables are most closely correlated? What does this suggest for the specification of regressions whose dependent variable is the bid-ask spread or the Amihud illiquidity ratio?
 - b. Estimate regressions where the dependent variable is the bid-ask spread, in six specifications that all include a constant, volatility, and *alternatively* the following other variables: (1) market capitalization, (2) trading volume, (3) log of market capitalization, (4) log of trading volume, (5) turnover rate (defined as $vo/ibnosh$); (6) turnover rate and log of closing price. Compare the explanatory power of these alternative specifications and, based on the models studied in this chapter, interpret the estimates so obtained. In particular, indicate the change in the bid-ask spread associated with a 1% increase in trading volume, according to your estimates.
 - c. Repeat the empirical analysis under (b) using the Amihud illiquidity measure as dependent variable. How do the results compare with those under (b)?
 - d. Would the theories studied in this chapter lead one to suppose that financial stocks should have been more illiquid than non-financial, other things being equal, in the estimation period of April 2009? Investigate by generating a financial sector dummy variable (*fin*) that takes the value 1 when the *gics* variable is equal to 40 (the code of the financial sector according to the Global Industry Classification Standard), and then by re-estimating specifications 4 and 6 described

under point (b) of this exercise with the addition of this dummy among the regressors. Comment on your results.

- e. The specification estimated in answering the previous question allows for the possibility that financial stocks may have been more illiquid than other stocks for given values of all the other explanatory variables. In fact, in the financial sector, the relationships of illiquidity with capitalization, trading volume, and price may not have been the same as in other sectors, in April 2009. Test this hypothesis within specification 6 under point (b) of this exercise (testing the equality of the coefficients of the explanatory variables first separately, and then jointly), and offer an interpretation of your results.